Appendix

The Appendix is organized as follows:

(1) Table 2 maps the meta-theoretical results in this paper to their Agda mechanization.
(2) Figure 9 presents fully static types, type well-formedness, and subtyping for $\lambda^{\alpha}_{\text{Ref}}$.
(3) Figure 10 presents gradual types, type well-formedness, and consistent subtyping for $\lambda^{\alpha,\star}_{\text{Ref}}$.
(4) Figure 11 presents terms and bidirectional typing rules for $\lambda^{\alpha,\star}_{\text{Ref}}$.
(5) Figure 12 defines code coercion sequence syntax, typing, and reduction semantics.
(6) Figure 13 defines meta coercion sequence syntax, typing, and normal forms.
(7) Figure 14 defines meta coercion sequence reduction semantics.
(8) Figure 15 defines well-formed typing contexts of $\text{CC}^{\alpha,\star}_{\text{Ref}}$.
(9) Figure 16 gives the typing rules for $\text{CC}^{\alpha,\star}_{\text{Ref}}$.
(10) Figure 17 defines well-typed runtime configurations for $\text{CC}^{\alpha,\star}_{\text{Ref}}$.
(11) Figure 18 presents the code reduction rules for $\text{CC}^{\alpha,\star}_{\text{Ref}}$.
(12) Figure 19 presents the meta reduction rules for $\text{CC}^{\alpha,\star}_{\text{Ref}}$.
(13) Figure 20 defines the coerce functions that generate coercions.
(14) Figure 21 defines elaboration from $\lambda^{\alpha,\star}_{\text{Ref}}$ to $\text{CC}^{\alpha,\star}_{\text{Ref}}$.
(15) Figure 22 defines the projection functions and inclusion relations used by preservation.
(16) Figure 23 shows a gradual quicksort program with potential catastrophic slowdown.
(17) Figure 24 defines meta hypercoercion syntax, typing, and composition.
(18) Figure 25 defines EC hypercoercion syntax, typing, and composition.
(19) Figure 26 defines code hypercoercion syntax, typing, and composition.
(20) Figure 27 defines the coerce functions that generate hypercoercions.
(21) Figure 28 defines the space-efficient reduction rules for $\text{CC}^{\alpha,\star}_{\text{Ref}}$ that use hypercoercions.
(22) Figure 29 defines adjacent cast judgments for code and meta terms.
(23) Figure 30 defines the adjacent cast judgment for heaps.
(24) Figure 31 defines hypercoercion height.
(25) Figure 32 defines hypercoercion size.
(26) Figure 33 defines cast height for code and meta terms.
(27) Figure 34 defines size, num, and ideal_size functions for code and meta terms.
(28) Figure 35 defines size and height functions for heaps.
(29) Figure 36 defines size and height functions for configurations.

Results	Agda identifiers	Agda files
Section 7: Type and scope safety		
Preservation for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$ (Lemma 7.1)	`preserve°`, `preserveᵐ`	`CCAlphaRefStarPreservation.agda`
Progress for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$ (Lemma 7.2)	`progress°`, `progressᵐ`	`CCAlphaRefStarProgress.agda`
Elaboration preserves types (Lemma 7.3)	`code-synth-preserves`, `code-check-preserves`, `meta-elab-preserves`	`ElaborationPreserves.agda`
Reification preserves types (Lemma 7.4)	`reify-preserves-types`	`ReifyPreserves.agda`
Reification is total for code values (Lemma 7.5)	`scope-safety-val`	`ScopeSafetyValue.agda`
Observation for code-typed values (Lemma 7.6)	`obs-code`	`CanonicalForms.agda`
Scope safety for $\lambda^{\alpha,\star}_{\mathrm{Ref}}$ (Theorem 7.7)	`scope-safety`	`ScopeSafety.agda`
Scope safety for $\lambda^{\alpha}_{\mathrm{Ref}}$ (Corollary 7.8)	`static-scope-safety`	`StaticScopeSafety.agda`
Section 8: Space efficiency		
Space consumption (Theorem 8.1)	`space-consumption-elab`	`SpaceEfficiency/SpaceEfficient.agda`
Elaboration efficient (Lemma 8.2)	`elab-efficient`, `elab-efficient°⇒`, `elab-efficient°⇐`	`SpaceEfficiency/Proofs/ElaborationEfficient.agda`
Preserve adjacent casts (Lemma 8.3)	`preserve-okᵐ`, `preserve-ok°`	`SpaceEfficiency/Proofs/PreserveOK.agda`
Preserve adjacent casts in heaps (Lemma 8.4)	`preserve-heap-values-okᵐ`, `preserve-heap-values-ok°`	`SpaceEfficiency/Proofs/PreserveOK.agda`
Composition height (Lemma 8.5)	`compose-height-code`, `compose-height-code-middle`, `compose-height-meta`, `compose-height-meta-middle`	`SpaceEfficiency/Proofs/HypercoercionCompositionBounded.agda`
Preserve height (Lemma 8.6)	`preserve-heightᵐ`, `preserve-height°`,	`SpaceEfficiency/Proofs/PreserveHeight.agda`
Size bounded by number of terms (Lemma 8.7)	`real-size-tsizeᵐ`, `real-size-tsize°`	`SpaceEfficiency/Proofs/RealSizeTermAndHeapSize.agda`
Number of terms bounded by ideal size (Lemma 8.8)	`size-OK`, `size°-OK`	`SpaceEfficiency/Proofs/SizeOK.agda`

Table 2. Mapping from meta-theoretical results to their Agda mechanization. File paths are relative to the root of the Agda development (the `Theories/` directory)

EC variables	α, β, γ		
ECs	e	$::=$	$\alpha \mid \varepsilon$
code types	T	$::=$	$\iota \mid T \to T$
meta types	A, B	$::=$	$\iota \mid A \to B \mid \mathtt{Ref}\ A \mid ``T''e \mid \forall\alpha.A \mid e{<:}e{\Rightarrow}A$
contexts	Γ	$::=$	$\emptyset \mid \Gamma, \alpha \mid \Gamma, e_1{<:}e_2 \mid \Gamma, x : A \mid \Gamma, (x : T)^{\alpha}$
subtype contexts	Θ	$::=$	$\emptyset \mid \Theta, e_1{<:}e_2$

$\boxed{\Gamma \vdash e}$

$$\frac{}{\Gamma \vdash \varepsilon} \qquad \frac{\alpha \in \Gamma}{\Gamma \vdash \alpha}$$

$\boxed{\Gamma \vdash A}$

$$\frac{}{\Gamma \vdash \iota} \qquad \frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \to B} \qquad \frac{\Gamma \vdash A}{\Gamma \vdash \mathtt{Ref}\ A}$$

$$\frac{\Gamma \vdash e}{\Gamma \vdash ``T''e} \qquad \frac{\Gamma, \alpha \vdash A}{\Gamma \vdash \forall\alpha.A} \qquad \frac{\Gamma \vdash e_1 \quad \Gamma \vdash e_2 \quad \Gamma \vdash A}{\Gamma \vdash e_1{<:}e_2{\Rightarrow}A}$$

$\boxed{\Theta \vdash e <: e}$

$$\frac{e_1 <: e_2 \in \Theta}{\Theta \vdash e_1 <: e_2} \qquad \frac{}{\Theta \vdash \varepsilon <: e} \qquad \frac{}{\Theta \vdash e <: e} \qquad \frac{\Theta \vdash e_1 <: e_2 \quad \Theta \vdash e_2 <: e_3}{\Theta \vdash e_1 <: e_3}$$

$\boxed{\Theta \vdash A <: B}$

$$\frac{}{\Theta \vdash \iota <: \iota} \qquad \frac{\Theta \vdash A' <: A \quad \Theta \vdash B <: B'}{\Theta \vdash A \to B <: A' \to B'} \qquad \frac{\Theta \vdash A' <: A \quad \Theta \vdash A <: A'}{\Theta \vdash \mathtt{Ref}\ A <: \mathtt{Ref}\ A'}$$

$$\frac{\Theta \vdash e_1 <: e_2}{\Theta \vdash ``T''e_1 <: ``T''e_2} \qquad \frac{\Theta \vdash A <: B}{\Theta \vdash \forall\alpha.A <: \forall\alpha.B} \qquad \frac{\Theta, e_1 <: e_2 \vdash A <: B}{\Theta \vdash (e_1{<:}e_2{\Rightarrow}A) <: (e_1{<:}e_2{\Rightarrow}B)}$$

Fig. 9. Fully static types, type well-formedness, and subtyping in $\lambda^{\alpha}_{\mathsf{Ref}}$

gradual ECs	$\hat{e}$	::=	$\star \mid e$
gradual code types	$\hat{T}$	::=	$\star \mid \iota \mid \hat{T} \to \hat{T}$
gradual meta types	$\hat{A}, \hat{B}$	::=	$\star \mid \iota \mid \hat{A} \to \hat{B} \mid \mathsf{Ref}\ \hat{A} \mid \grave{}\grave{}\hat{T}\,'\,'\hat{e} \mid \forall \boldsymbol{\alpha}.\, \hat{A} \mid e{<}{:}e{\Rightarrow}\hat{A}$
gradual contexts	Γ	::=	$\emptyset \mid \Gamma, \boldsymbol{\alpha} \mid \Gamma, e_1{<}{:}e_2 \mid \Gamma, x : \hat{A} \mid \Gamma, (x : T)^{\boldsymbol{\alpha}}$

$$\boxed{\Gamma \vdash \hat{e}} \qquad \boxed{\Gamma \vdash \hat{A}}$$

$$\frac{}{\Gamma \vdash \star} \qquad \frac{}{\Gamma \vdash \varepsilon} \qquad \frac{\alpha \in \Gamma}{\Gamma \vdash \alpha} \qquad \frac{}{\Gamma \vdash \star} \qquad \frac{}{\Gamma \vdash \iota} \qquad \frac{\Gamma \vdash \hat{A} \quad \Gamma \vdash \hat{B}}{\Gamma \vdash \hat{A} \to \hat{B}} \qquad \frac{\Gamma \vdash \hat{A}}{\Gamma \vdash \mathsf{Ref}\ \hat{A}}$$

$$\frac{\Gamma \vdash \hat{e}}{\Gamma \vdash \grave{}\grave{}\hat{T}\,'\,'\hat{e}} \qquad \frac{\Gamma, \alpha \vdash \hat{A}}{\Gamma \vdash \forall \alpha.\, \hat{A}} \qquad \frac{\Gamma \vdash e_1 \quad \Gamma \vdash e_2 \quad \Gamma \vdash \hat{A}}{\Gamma \vdash e_1{<}{:}e_2{\Rightarrow}\hat{A}}$$

$$\boxed{\Theta \vdash \hat{e} \lesssim \hat{e}} \qquad \boxed{\Theta \vdash \hat{T} \lesssim \hat{T}}$$

$$\frac{}{\Theta \vdash \hat{e} \lesssim \star} \qquad \frac{}{\Theta \vdash \star \lesssim \hat{e}} \qquad \frac{\Theta \vdash e_1 <: e_2}{\Theta \vdash e_1 \lesssim e_2}$$

$$\frac{}{\Theta \vdash \hat{T} \lesssim \star} \qquad \frac{}{\Theta \vdash \star \lesssim \hat{T}} \qquad \frac{}{\Theta \vdash \iota \lesssim \iota} \qquad \frac{\Theta \vdash \hat{T}_1' \lesssim \hat{T}_1 \quad \Theta \vdash \hat{T}_2 \lesssim \hat{T}_2'}{\Theta \vdash \hat{T}_1 \to \hat{T}_2 \lesssim \hat{T}_1' \to \hat{T}_2'}$$

$$\boxed{\Theta \vdash \hat{A} \lesssim \hat{B}}$$

$$\frac{}{\Theta \vdash \hat{A} \lesssim \star} \qquad \frac{}{\Theta \vdash \star \lesssim \hat{B}} \qquad \frac{}{\Theta \vdash \iota \lesssim \iota} \qquad \frac{\Theta \vdash \hat{A}' \lesssim \hat{A} \quad \Theta \vdash \hat{B} \lesssim \hat{B}'}{\Theta \vdash \hat{A} \to \hat{B} \lesssim \hat{A}' \to \hat{B}'}$$

$$\frac{\Theta \vdash \hat{A}' \lesssim \hat{A} \quad \Theta \vdash \hat{A} \lesssim \hat{A}'}{\Theta \vdash \mathsf{Ref}\ \hat{A} \lesssim \mathsf{Ref}\ \hat{A}'} \qquad \frac{\Theta \vdash \hat{T}_1 \lesssim \hat{T}_2 \quad \Theta \vdash \hat{e}_1 \lesssim \hat{e}_2}{\Theta \vdash \grave{}\grave{}\hat{T}_1\,'\,'\hat{e}_1 \lesssim \grave{}\grave{}\hat{T}_2\,'\,'\hat{e}_2}$$

$$\frac{\Theta \vdash \hat{A} \lesssim \hat{B}}{\Theta \vdash \forall \alpha.\, \hat{A} \lesssim \forall \alpha.\, \hat{B}} \qquad \frac{\Theta, e_1{<}{:}e_2 \vdash \hat{A} \lesssim \hat{B}}{\Theta \vdash (e_1{<}{:}e_2{\Rightarrow}\hat{A}) \lesssim (e_1{<}{:}e_2{\Rightarrow}\hat{B})}$$

Fig. 10. Gradual types, type well-formedness, and consistent subtyping for $\lambda^{\alpha,\star}_{\mathsf{Ref}}$

$$\begin{array}{lrcl} \text{code terms} & L^\circ, M^\circ, N^\circ & ::= & k \mid x \mid \lambda(x{:}T)^{\alpha}.\, M^\circ \mid (L^\circ\ M^\circ)^{\ell} \mid {\sim}^{\ell} M \\ \text{meta terms} & L, M, N & ::= & k \mid x \mid \lambda x{:}\hat{A}.\, M \mid (L\ M)^{\ell} \mid \Lambda\alpha.\, M \mid M[e]^{\ell} \mid e_1{<:}e_2 \Rightarrow M \mid M\bullet^{\ell} \\ & & \mid & \grave{}\grave{}M^\circ{}'{}'e \mid \texttt{ref}\ M \mid L :=^{\ell} M \mid !^{\ell} M \end{array}$$

$\boxed{\Gamma; e \vdash M^\circ \Rightarrow T}$

$$\frac{k : \iota}{\Gamma; e \vdash k \Rightarrow \iota}\ (\vdash \textit{code-const}) \qquad \frac{(x : T)^{\alpha} \in \Gamma \qquad \mathrm{Sub}(\Gamma) \vdash \alpha <: e}{\Gamma; e \vdash x \Rightarrow T}\ (\vdash \textit{code-var})$$

$$\frac{\Gamma, \alpha, e{<:}\alpha, (x : T_1)^{\alpha}; \alpha \vdash N^\circ \Rightarrow T_2 \qquad \alpha \notin \Gamma}{\Gamma; e \vdash \lambda(x{:}T_1)^{\alpha}.\, N^\circ \Rightarrow T_1 \to T_2}\ (\vdash \textit{code-abs}) \qquad \frac{\Gamma; e \vdash L^\circ \Rightarrow T_1 \to T_2 \qquad \Gamma; e \vdash M^\circ \Leftarrow T_1}{\Gamma; e \vdash (L^\circ\ M^\circ)^{\ell} \Rightarrow T_2}\ (\vdash \textit{code-app})$$

$\boxed{\Gamma; e \vdash M^\circ \Leftarrow T}$

$$\frac{\Gamma \vdash M : \hat{A} \qquad \mathrm{Sub}(\Gamma) \vdash \hat{A} \lesssim \grave{}\grave{}T{}'{}'e}{\Gamma; e \vdash {\sim}^{\ell} M \Leftarrow T}\ (\vdash \textit{splice}) \qquad \frac{\Gamma; e \vdash M^\circ \Rightarrow T}{\Gamma; e \vdash M^\circ \Leftarrow T}\ (\vdash \textit{check})$$

$\boxed{\Gamma \vdash M : \hat{A}}$

$$\frac{k : \iota}{\Gamma \vdash k : \iota}\ (\vdash \textit{meta-const}) \qquad \frac{x : \hat{A} \in \Gamma}{\Gamma \vdash x : \hat{A}}\ (\vdash \textit{meta-var}) \qquad \frac{\Gamma \vdash \hat{A} \qquad \Gamma, x : \hat{A} \vdash M : \hat{B}}{\Gamma \vdash \lambda x{:}\hat{A}.\, M : \hat{A} \to \hat{B}}\ (\vdash \textit{meta-abs})$$

$$\frac{\Gamma \vdash L : \hat{A}_1 \to \hat{A}_2 \qquad \Gamma \vdash M : \hat{B} \qquad \mathrm{Sub}(\Gamma) \vdash \hat{B} \lesssim \hat{A}_1}{\Gamma \vdash (L\ M)^{\ell} : \hat{A}_2}\ (\vdash \textit{meta-app}) \qquad \frac{\Gamma \vdash L : \star \qquad \Gamma \vdash M : \hat{A}}{\Gamma \vdash (L\ M)^{\ell} : \star}\ (\vdash \textit{meta-app-}\star) \qquad \frac{\Gamma, \alpha \vdash M : \hat{A} \qquad \alpha \notin \Gamma}{\Gamma \vdash \Lambda\alpha.\, M : \forall\alpha.\, \hat{A}}\ (\vdash \textit{ec-abs})$$

$$\frac{\Gamma \vdash e \qquad \Gamma \vdash M : \forall\alpha.\, \hat{A}}{\Gamma \vdash M[e]^{\ell} : \hat{A}[\alpha := e]}\ (\vdash \textit{ec-app}) \qquad \frac{\Gamma \vdash e \qquad \Gamma \vdash M : \star}{\Gamma \vdash M[e]^{\ell} : \star}\ (\vdash \textit{ec-app-}\star) \qquad \frac{\Gamma \vdash e_1 \qquad \Gamma \vdash e_2 \qquad \Gamma, e_1{<:}e_2 \vdash M : \hat{A}}{\Gamma \vdash e_1{<:}e_2 \Rightarrow M : e_1{<:}e_2 \Rightarrow \hat{A}}\ (\vdash \textit{sub-intro})$$

$$\frac{\mathrm{Sub}(\Gamma) \vdash e_1 <: e_2 \qquad \Gamma \vdash M : e_1{<:}e_2 \Rightarrow \hat{A}}{\Gamma \vdash M\bullet^{\ell} : \hat{A}}\ (\vdash \textit{sub-elim}) \qquad \frac{\Gamma \vdash M : \star}{\Gamma \vdash M\bullet^{\ell} : \star}\ (\vdash \textit{sub-elim-}\star) \qquad \frac{\Gamma \vdash e \qquad \Gamma; e \vdash M^\circ \Rightarrow T}{\Gamma \vdash \grave{}\grave{}M^\circ{}'{}'e : \grave{}\grave{}T{}'{}'e}\ (\vdash \textit{quote})$$

$$\frac{\Gamma \vdash M : \hat{A}}{\Gamma \vdash \texttt{ref}\ M : \texttt{Ref}\ \hat{A}}\ (\vdash \textit{ref}) \qquad \frac{\Gamma \vdash L : \texttt{Ref}\ \hat{A} \qquad \Gamma \vdash M : \hat{B} \qquad \mathrm{Sub}(\Gamma) \vdash \hat{B} \lesssim \hat{A}}{\Gamma \vdash L :=^{\ell} M : \texttt{Unit}}\ (\vdash \textit{assign}) \qquad \frac{\Gamma \vdash L : \star \qquad \Gamma \vdash M : \hat{A}}{\Gamma \vdash L :=^{\ell} M : \texttt{Unit}}\ (\vdash \textit{assign-}\star)$$

$$\frac{\Gamma \vdash M : \texttt{Ref}\ \hat{A}}{\Gamma \vdash !^{\ell} M : \hat{A}}\ (\vdash \textit{deref}) \qquad \frac{\Gamma \vdash M : \star}{\Gamma \vdash !^{\ell} M : \star}\ (\vdash \textit{deref-}\star)$$

Fig. 11. Terms and bidirectional typing rules for $\lambda^{\alpha,\star}_{\mathrm{Ref}}$

New rules (compared to $\lambda^{\alpha}_{\mathrm{Ref}}$ of Figure 2) are highlighted in green. The premises changes from subtyping to consistent subtyping are highlighted in blue. Orange text (ℓ) indicates blame labels.

$$
\begin{array}{lll}
\text{ground code types} & G^\circ, H^\circ & ::= \iota \mid \star \to \star \\
\text{single code coercions} & c^\circ, d^\circ & ::= G^\circ! \mid G^\circ?^\ell \mid \bar{c}^\circ \to \bar{d}^\circ \mid \bot^\ell \\
\text{code coercion sequences} & \bar{c}^\circ, \bar{d}^\circ & ::= [] \mid c^\circ :: \bar{c}^\circ \\
\text{single normal forms} & c^\circ_{\text{NF}}, d^\circ_{\text{NF}} & ::= G^\circ! \mid G^\circ?^\ell \mid \bot^\ell \mid (\bar{c}^\circ_{\text{NF}} \to \bar{d}^\circ_{\text{NF}})^\dagger \\
\text{normal forms} & \bar{c}^\circ_{\text{NF}}, \bar{d}^\circ_{\text{NF}} & ::= [] \mid [c^\circ_{\text{NF}}] \mid (c^\circ_{\text{NF}} :: d^\circ_{\text{NF}} :: \bar{d}^\circ_{\text{NF}})^\ddagger
\end{array}
$$

$^\dagger \neg(\mathsf{Id}(\bar{c}^\circ_{\text{NF}}) \wedge \mathsf{Id}(\bar{d}^\circ_{\text{NF}})), \neg\mathsf{Blame}(\bar{c}^\circ_{\text{NF}}), \neg\mathsf{Blame}(\bar{d}^\circ_{\text{NF}})$; $\quad ^\ddagger \mathsf{IrredPair}(c^\circ_{\text{NF}}, d^\circ_{\text{NF}})$.

$$\mathsf{Id}(\bar{c}^\circ) \triangleq \bar{c}^\circ = [] \qquad \mathsf{Blame}(\bar{c}^\circ) \triangleq \exists \ell.\ \bar{c}^\circ = [\bot^\ell]$$

$\boxed{\vdash c^\circ : \hat{T}_1 \Rightarrow \hat{T}_2}$

$$\frac{}{\vdash G^\circ! : G^\circ \Rightarrow \star}\ (\vdash !) \qquad \frac{}{\vdash G^\circ?^\ell : \star \Rightarrow G^\circ}\ (\vdash ?)$$

$$\frac{\vdash \bar{c}^\circ : \hat{T}_3 \Rightarrow \hat{T}_1 \qquad \vdash \bar{d}^\circ : \hat{T}_2 \Rightarrow \hat{T}_4}{\vdash \bar{c}^\circ \to \bar{d}^\circ : (\hat{T}_1 \to \hat{T}_2) \Rightarrow (\hat{T}_3 \to \hat{T}_4)}\ (\vdash \mathit{fun}) \qquad \frac{}{\vdash \bot^\ell : \hat{T}_1 \Rightarrow \hat{T}_2}\ (\vdash \bot)$$

$\boxed{\vdash \bar{c}^\circ : \hat{T}_1 \Rightarrow \hat{T}_2}$

$$\frac{}{\vdash [] : \hat{T} \Rightarrow \hat{T}}\ (\vdash \mathit{empty}) \qquad \frac{\vdash c^\circ : \hat{T}_1 \Rightarrow \hat{T}_2 \qquad \vdash \bar{c}^\circ : \hat{T}_2 \Rightarrow \hat{T}_3}{\vdash c^\circ :: \bar{c}^\circ : \hat{T}_1 \Rightarrow \hat{T}_3}\ (\vdash \mathit{cons})$$

$\boxed{c^\circ \longrightarrow \bar{d}^\circ}$

$$\frac{}{[] \to [] \longrightarrow []}\ (\mathit{id\text{-}fun})$$

$$\frac{\bar{c}^\circ_1 \longrightarrow \bar{d}^\circ_1}{\bar{c}^\circ_1 \to \bar{c}^\circ_2 \longrightarrow [\bar{d}^\circ_1 \to \bar{c}^\circ_2]}\ (\xi\text{-}\mathit{fun\text{-}dom}) \qquad \frac{\neg\mathsf{Blame}(\bar{c}^\circ_{\text{NF}}) \qquad \bar{c}^\circ \longrightarrow \bar{d}^\circ}{\bar{c}^\circ_{\text{NF}} \to \bar{c}^\circ \longrightarrow [\bar{c}^\circ_{\text{NF}} \to \bar{d}^\circ]}\ (\xi\text{-}\mathit{fun\text{-}cod})$$

$$\frac{}{[\bot^\ell] \to \bar{c}^\circ \longrightarrow [\bot^\ell]}\ (\xi\text{-}\bot\text{-}\mathit{fun\text{-}dom}) \qquad \frac{\neg\mathsf{Blame}(\bar{c}^\circ_{\text{NF}})}{\bar{c}^\circ_{\text{NF}} \to [\bot^\ell] \longrightarrow [\bot^\ell]}\ (\xi\text{-}\bot\text{-}\mathit{fun\text{-}cod})$$

$\boxed{c^\circ_1; c^\circ_2 \longrightarrow \bar{d}^\circ}$

$$\frac{}{G^\circ!; G^\circ?^\ell \longrightarrow []}\ (\mathit{collapse}) \qquad \frac{G^\circ \neq H^\circ}{G^\circ!; H^\circ?^\ell \longrightarrow [\bot^\ell]}\ (\mathit{collide})$$

$$\frac{}{(\bar{c}^\circ_1 \to \bar{c}^\circ_2); (\bar{d}^\circ_1 \to \bar{d}^\circ_2) \longrightarrow [(\bar{d}^\circ_1 +\!\!+ \bar{c}^\circ_1) \to (\bar{c}^\circ_2 +\!\!+ \bar{d}^\circ_2)]}\ (\mathit{seq\text{-}fun})$$

$$\frac{}{\bot^\ell; c^\circ \longrightarrow [\bot^\ell]}\ (\bot\text{-}\mathit{left}) \qquad \frac{}{G^\circ!; \bot^\ell \longrightarrow [\bot^\ell]}\ (\bot\text{-}\mathit{right\text{-}!}) \qquad \frac{}{(\bar{c}^\circ_1 \to \bar{c}^\circ_2); \bot^\ell \longrightarrow [\bot^\ell]}\ (\bot\text{-}\mathit{right\text{-}fun})$$

$\boxed{\bar{c}^\circ \longrightarrow \bar{d}^\circ}$

$$\frac{c^\circ \longrightarrow \bar{d}^\circ}{c^\circ :: \bar{c}^\circ \longrightarrow \bar{d}^\circ +\!\!+ \bar{c}^\circ}\ (\mathit{reduce\text{-}head}) \qquad \frac{c^\circ_{\text{NF}}; c^\circ \longrightarrow \bar{d}^\circ}{c^\circ_{\text{NF}} :: c^\circ :: \bar{c}^\circ \longrightarrow \bar{d}^\circ +\!\!+ \bar{c}^\circ}\ (\mathit{reduce\text{-}neighbors})$$

$$\frac{\mathsf{IrredPair}(c^\circ_{\text{NF}}, d^\circ_{\text{NF}}) \qquad d^\circ_{\text{NF}} :: \bar{c}^\circ \longrightarrow \bar{d}^\circ}{c^\circ_{\text{NF}} :: d^\circ_{\text{NF}} :: \bar{c}^\circ \longrightarrow c^\circ_{\text{NF}} :: \bar{d}^\circ}\ (\mathit{reduce\text{-}tail})$$

$\boxed{\mathsf{IrredPair}(c^\circ, d^\circ)}$

$$\frac{\mathsf{IrredPair}(G^\circ?^\ell, G^\circ!)}{\mathsf{IrredPair}((\star \to \star)?^\ell, \bar{c}^\circ_1 \to \bar{c}^\circ_2)} \qquad \frac{\mathsf{IrredPair}(G^\circ?^{\ell_1}, \bot^{\ell_2})}{\mathsf{IrredPair}(\bar{c}^\circ_1 \to \bar{c}^\circ_2, (\star \to \star)!)}$$

Fig. 12. Code coercion sequences: syntax, typing, and reduction semantics

ground meta types	G, H	::=	$\grave{}\grave{}\star\acute{}\acute{}\star \mid \forall\alpha.\star \mid e{<:}e{\Rightarrow}\star \mid \iota \mid \star \to \star \mid \mathtt{Ref}\ \star$
single meta coercions	c, d	::=	$G! \mid G?^{\ell} \mid \bot^{\ell} \mid \bar{c} \to \bar{d} \mid \grave{}\grave{}\bar{c}^{\circ}\acute{}\acute{}\bar{c}^{e} \mid \mathtt{Ref}\ \bar{c}\ \bar{d} \mid \forall\alpha.\bar{c} \mid e_1{<:}e_2{\Rightarrow}\bar{c}$
meta coercion sequences	$\bar{c}, \bar{d}$	::=	$[] \mid c :: \bar{c}$
single normal forms	$c_{\mathrm{NF}}, d_{\mathrm{NF}}$	::=	$G! \mid G?^{\ell} \mid \bot^{\ell} \mid (\bar{c}_{\mathrm{NF}} \to \bar{d}_{\mathrm{NF}})^{\dagger_1} \mid (\grave{}\grave{}\bar{c}^{\circ}_{\mathrm{NF}}\acute{}\acute{}\bar{c}^{e}_{\mathrm{NF}})^{\dagger_2} \mid (\mathtt{Ref}\ \bar{c}_{\mathrm{NF}}\ \bar{d}_{\mathrm{NF}})^{\dagger_3}$
		|	$(\forall\alpha.\bar{c}_{\mathrm{NF}})^{\dagger_4} \mid (e_1{<:}e_2{\Rightarrow}\bar{c}_{\mathrm{NF}})^{\dagger_5}$
normal forms	$\bar{c}_{\mathrm{NF}}, \bar{d}_{\mathrm{NF}}$	::=	$[] \mid [c_{\mathrm{NF}}] \mid (c_{\mathrm{NF}} :: \bar{d}_{\mathrm{NF}})^{\dagger_6}$

$^{\dagger_1}\neg(\mathsf{Id}(\bar{c}_{\mathrm{NF}}) \wedge \mathsf{Id}(\bar{d}_{\mathrm{NF}}))$; $^{\dagger_2}\neg(\mathsf{Id}(\bar{c}^{\circ}_{\mathrm{NF}}) \wedge \mathsf{Id}(\bar{c}^{e}_{\mathrm{NF}})), \neg\mathsf{Blame}(\bar{c}^{\circ}_{\mathrm{NF}}), \neg\mathsf{Blame}(\bar{c}^{e}_{\mathrm{NF}})$; $^{\dagger_3}\neg(\mathsf{Id}(\bar{c}_{\mathrm{NF}}) \wedge \mathsf{Id}(\bar{d}_{\mathrm{NF}}))$;
$^{\dagger_4}\neg\mathsf{Id}(\bar{c}_{\mathrm{NF}})$; $^{\dagger_5}\neg\mathsf{Id}(\bar{c}_{\mathrm{NF}})$; $^{\dagger_6}\mathsf{IrredPair}(c_{\mathrm{NF}}, d_{\mathrm{NF}})$.

$$\mathsf{Id}(\bar{c}) \triangleq \bar{c} = [] \qquad \mathsf{Blame}(c) \triangleq \exists\ell.\ c = \bot^{\ell} \qquad \mathsf{Proj}(c) \triangleq \exists G, \ell.\ c = G?^{\ell}$$

$\boxed{\Gamma \vdash c : \hat{A} \Rightarrow \hat{B}}$

$$\frac{\Gamma \vdash G}{\Gamma \vdash G! : G \Rightarrow \star}\ (\vdash !) \qquad \frac{\Gamma \vdash G}{\Gamma \vdash G?^{\ell} : \star \Rightarrow G}\ (\vdash ?) \qquad \frac{\Gamma \vdash \hat{A} \quad \Gamma \vdash \hat{B}}{\Gamma \vdash \bot^{\ell} : \hat{A} \Rightarrow \hat{B}}\ (\vdash \bot)$$

$$\frac{\Gamma \vdash \bar{c} : \hat{A}' \Rightarrow \hat{A} \quad \Gamma \vdash \bar{d} : \hat{B} \Rightarrow \hat{B}'}{\Gamma \vdash \bar{c} \to \bar{d} : (\hat{A} \to \hat{B}) \Rightarrow (\hat{A}' \to \hat{B}')}\ (\vdash fun) \qquad \frac{\vdash \bar{c}^{\circ} : \hat{T}_1 \Rightarrow \hat{T}_2 \quad \Gamma \vdash \bar{c}^{e} : \hat{e}_1 \Rightarrow \hat{e}_2}{\Gamma \vdash \grave{}\grave{}\bar{c}^{\circ}\acute{}\acute{}\bar{c}^{e} : \grave{}\grave{}\hat{T}_1\acute{}\acute{}\hat{e}_1 \Rightarrow \grave{}\grave{}\hat{T}_2\acute{}\acute{}\hat{e}_2}\ (\vdash code)$$

$$\frac{\Gamma \vdash \bar{c} : \hat{B} \Rightarrow \hat{A} \quad \Gamma \vdash \bar{d} : \hat{A} \Rightarrow \hat{B}}{\Gamma \vdash \mathtt{Ref}\ \bar{c}\ \bar{d} : \mathtt{Ref}\ \hat{A} \Rightarrow \mathtt{Ref}\ \hat{B}}\ (\vdash ref) \qquad \frac{\Gamma, \alpha \vdash \bar{c} : \hat{A} \Rightarrow \hat{B}}{\Gamma \vdash \forall\alpha.\bar{c} : \forall\alpha.\hat{A} \Rightarrow \forall\alpha.\hat{B}}\ (\vdash \forall)$$

$$\frac{\Gamma \vdash e_1 \quad \Gamma \vdash e_2 \quad \Gamma, e_1{<:}e_2 \vdash \bar{c} : \hat{A} \Rightarrow \hat{B}}{\Gamma \vdash e_1{<:}e_2{\Rightarrow}\bar{c} : (e_1{<:}e_2{\Rightarrow}\hat{A}) \Rightarrow (e_1{<:}e_2{\Rightarrow}\hat{B})}\ (\vdash sub)$$

$\boxed{\Gamma \vdash \bar{c} : \hat{A} \Rightarrow \hat{B}}$

$$\frac{\Gamma \vdash \hat{A}}{\Gamma \vdash [] : \hat{A} \Rightarrow \hat{A}}\ (\vdash empty) \qquad \frac{\Gamma \vdash c : \hat{A} \Rightarrow \hat{B} \quad \Gamma \vdash \bar{c} : \hat{B} \Rightarrow \hat{C}}{\Gamma \vdash c :: \bar{c} : \hat{A} \Rightarrow \hat{C}}\ (\vdash cons)$$

$\boxed{\mathsf{IrredPair}(c, d)}$

$$\frac{}{\mathsf{IrredPair}(G?^{\ell}, G!)} \qquad \frac{}{\mathsf{IrredPair}(G?^{\ell}, \bot^{\ell'})}$$

$$\frac{}{\mathsf{IrredPair}((\star \to \star)?^{\ell}, \bar{c} \to \bar{d})} \qquad \frac{}{\mathsf{IrredPair}(\bar{c} \to \bar{d}, (\star \to \star)!)}$$

$$\frac{}{\mathsf{IrredPair}((\grave{}\grave{}\star\acute{}\acute{}\star)?^{\ell}, \grave{}\grave{}\bar{c}^{\circ}\acute{}\acute{}\bar{c}^{e})} \qquad \frac{}{\mathsf{IrredPair}(\grave{}\grave{}\bar{c}^{\circ}\acute{}\acute{}\bar{c}^{e}, (\grave{}\grave{}\star\acute{}\acute{}\star)!)}$$

$$\frac{}{\mathsf{IrredPair}((\mathtt{Ref}\ \star)?^{\ell}, \mathtt{Ref}\ \bar{c}\ \bar{d})} \qquad \frac{}{\mathsf{IrredPair}(\mathtt{Ref}\ \bar{c}\ \bar{d}, (\mathtt{Ref}\ \star)!)}$$

$$\frac{}{\mathsf{IrredPair}((\forall\alpha.\star)?^{\ell}, \forall\alpha.\bar{c})} \qquad \frac{}{\mathsf{IrredPair}(\forall\alpha.\bar{c}, (\forall\alpha.\star)!)}$$

$$\frac{}{\mathsf{IrredPair}((e_1{<:}e_2{\Rightarrow}\star)?^{\ell}, e_1{<:}e_2{\Rightarrow}\bar{c})} \qquad \frac{}{\mathsf{IrredPair}(e_1{<:}e_2{\Rightarrow}\bar{c}, (e_1{<:}e_2{\Rightarrow}\star)!)}$$

Fig. 13. Meta coercion sequences: syntax and typing, and normal forms

$\boxed{\Theta \vdash c \longrightarrow \bar{d}}$

$$\frac{}{\Theta \vdash [] \to [] \longrightarrow []}\ (id\text{-}fun) \qquad \frac{}{\Theta \vdash \text{``}[]\text{''}[] \longrightarrow []}\ (id\text{-}code) \qquad \frac{}{\Theta \vdash \texttt{Ref}\ []\ [] \longrightarrow []}\ (id\text{-}ref)$$

$$\frac{}{\Theta \vdash \forall\alpha.[] \longrightarrow []}\ (id\text{-}forall) \qquad \frac{}{\Theta \vdash e_1 <: e_2 \Rightarrow [] \longrightarrow []}\ (id\text{-}sub)$$

$$\frac{\Theta \vdash \bar{c}_1 \longrightarrow \bar{d}_1}{\Theta \vdash \bar{c}_1 \to \bar{c}_2 \longrightarrow [\bar{d}_1 \to \bar{c}_2]}\ (\xi\text{-}fun\text{-}dom) \qquad \frac{\Theta \vdash \bar{c} \longrightarrow \bar{d}}{\Theta \vdash \bar{c}_{\text{NF}} \to \bar{c} \longrightarrow [\bar{c}_{\text{NF}} \to \bar{d}]}\ (\xi\text{-}fun\text{-}cod)$$

$$\frac{\Theta \vdash \bar{c}_1 \longrightarrow \bar{d}_1}{\Theta \vdash \texttt{Ref}\ \bar{c}_1\ \bar{c}_2 \longrightarrow [\texttt{Ref}\ \bar{d}_1\ \bar{c}_2]}\ (\xi\text{-}ref\text{-}in) \qquad \frac{\Theta \vdash \bar{c} \longrightarrow \bar{d}}{\Theta \vdash \texttt{Ref}\ \bar{c}_{\text{NF}}\ \bar{c} \longrightarrow [\texttt{Ref}\ \bar{c}_{\text{NF}}\ \bar{d}]}\ (\xi\text{-}ref\text{-}out)$$

$$\frac{\Theta \vdash \bar{c} \longrightarrow \bar{d}}{\Theta \vdash \forall\alpha.\bar{c} \longrightarrow [\forall\alpha.\bar{d}]}\ (\xi\text{-}forall) \qquad \frac{\Theta, e_1 <: e_2 \vdash \bar{c} \longrightarrow \bar{d}}{\Theta \vdash e_1 <: e_2 \Rightarrow \bar{c} \longrightarrow [e_1 <: e_2 \Rightarrow \bar{d}]}\ (\xi\text{-}sub)$$

$$\frac{\bar{c}^{\circ} \longrightarrow \bar{d}^{\circ}}{\Theta \vdash \text{``}\bar{c}^{\circ}\text{''}\bar{c}^{e} \longrightarrow [\text{``}\bar{d}^{\circ}\text{''}\bar{c}^{e}]}\ (\xi\text{-}code) \qquad \frac{\neg\text{Blame}(\bar{c}^{\circ}_{\text{NF}}) \qquad \Theta \vdash \bar{c}^{e} \longrightarrow \bar{d}^{e}}{\Theta \vdash \text{``}\bar{c}^{\circ}_{\text{NF}}\text{''}\bar{c}^{e} \longrightarrow [\text{``}\bar{c}^{\circ}_{\text{NF}}\text{''}\bar{d}^{e}]}\ (\xi\text{-}code\text{-}ec)$$

$$\frac{}{\Theta \vdash \text{``}[\bot^{\ell}]\text{''}\bar{c}^{e} \longrightarrow [\bot^{\ell}]}\ (\xi\text{-}code\text{-}\bot) \qquad \frac{\neg\text{Blame}(\bar{c}^{\circ}_{\text{NF}})}{\Theta \vdash \text{``}\bar{c}^{\circ}_{\text{NF}}\text{''}[\bot^{\ell}] \longrightarrow [\bot^{\ell}]}\ (\xi\text{-}code\text{-}ec\text{-}\bot)$$

$\boxed{\Theta \vdash c_1; c_2 \longrightarrow \bar{d}}$

$$\frac{}{\Theta \vdash G!; G?^{\ell} \longrightarrow []}\ (collapse) \qquad \frac{G \neq H}{\Theta \vdash G!; H?^{\ell} \longrightarrow [\bot^{\ell}]}\ (collide)$$

$$\frac{}{\Theta \vdash (\bar{c}_1 \to \bar{c}_2); (\bar{d}_1 \to \bar{d}_2) \longrightarrow [(\bar{d}_1 +\!\!+ \bar{c}_1) \to (\bar{c}_2 +\!\!+ \bar{d}_2)]}\ (seq\text{-}fun)$$

$$\frac{}{\Theta \vdash \texttt{Ref}\ \bar{c}_1\ \bar{c}_2; \texttt{Ref}\ \bar{d}_1\ \bar{d}_2 \longrightarrow [\texttt{Ref}\ (\bar{d}_1 +\!\!+ \bar{c}_1)\ (\bar{c}_2 +\!\!+ \bar{d}_2)]}\ (seq\text{-}ref)$$

$$\frac{}{\Theta \vdash \text{``}\bar{c}^{\circ}\text{''}\bar{c}^{e}; \text{``}\bar{d}^{\circ}\text{''}\bar{d}^{e} \longrightarrow [\text{``}\bar{c}^{\circ} +\!\!+ \bar{d}^{\circ}\text{''}(\bar{c}^{e} +\!\!+ \bar{d}^{e})]}\ (seq\text{-}code)$$

$$\frac{}{\Theta \vdash (\forall\alpha.\bar{c}); (\forall\alpha.\bar{d}) \longrightarrow [\forall\alpha.(\bar{c} +\!\!+ \bar{d})]}\ (seq\text{-}forall)$$

$$\frac{}{\Theta \vdash (e_1 <: e_2 \Rightarrow \bar{c}); (e_1 <: e_2 \Rightarrow \bar{d}) \longrightarrow [e_1 <: e_2 \Rightarrow (\bar{c} +\!\!+ \bar{d})]}\ (seq\text{-}sub)$$

$$\frac{}{\Theta \vdash \bot^{\ell}; c \longrightarrow [\bot^{\ell}]}\ (\bot\text{-}left) \qquad \frac{\neg\text{Blame}(c) \qquad \neg\text{Proj}(c)}{\Theta \vdash c; \bot^{\ell} \longrightarrow [\bot^{\ell}]}\ (\bot\text{-}right)$$

$\boxed{\Theta \vdash \bar{c} \longrightarrow \bar{d}}$

$$\frac{\Theta \vdash c \longrightarrow \bar{d}}{\Theta \vdash c :: \bar{c} \longrightarrow \bar{d} +\!\!+ \bar{c}}\ (reduce\text{-}head) \qquad \frac{\Theta \vdash c_{\text{NF}}; c \longrightarrow \bar{d}}{\Theta \vdash c_{\text{NF}} :: c :: \bar{c} \longrightarrow \bar{d} +\!\!+ \bar{c}}\ (reduce\text{-}neighbors)$$

$$\frac{\text{IrredPair}(c_{\text{NF}}, d_{\text{NF}}) \qquad \Theta \vdash d_{\text{NF}} :: \bar{c} \longrightarrow \bar{d}}{\Theta \vdash c_{\text{NF}} :: d_{\text{NF}} :: \bar{c} \longrightarrow c_{\text{NF}}; \bar{d}}\ (reduce\text{-}tail)$$

Fig. 14. Meta coercion sequences: reduction semantics

$\boxed{\vdash \Gamma}$

$$\frac{}{\vdash \emptyset}\ \textit{(wf-empty)} \qquad \frac{\vdash \Gamma \qquad \Gamma \vdash \hat{A}}{\vdash \Gamma, x : \hat{A}}\ \textit{(wf-meta-var)} \qquad \frac{\vdash \Gamma \qquad \alpha \notin \Gamma}{\vdash \Gamma, \alpha_{\text{lex}}}\ \textit{(wf-ec-lex)} \qquad \frac{\vdash \Gamma \qquad \alpha \notin \Gamma}{\vdash \Gamma, \alpha_{\text{glob}}}\ \textit{(wf-ec-glob)}$$

$$\frac{\vdash \Gamma \qquad \Gamma \vdash e_1 \qquad \Gamma \vdash e_2}{\vdash \Gamma, (e_1 <: e_2)_{\text{lex}}}\ \textit{(wf-sub-lex)} \qquad \frac{\vdash \Gamma \qquad \Gamma \vdash e \qquad \Gamma \vdash \alpha \qquad \alpha_{\text{glob}} \in \Gamma \qquad \forall e'.\ (e' <: \alpha)_{\text{glob}} \notin \Gamma}{\vdash \Gamma, (e <: \alpha)_{\text{glob}}}\ \textit{(wf-sub-glob)}$$

$$\frac{\vdash \Gamma \qquad \Gamma \vdash \alpha}{\vdash \Gamma, (x{:}T)^{\alpha}_{\text{lex}}}\ \textit{(wf-code-var-lex)} \qquad \frac{\vdash \Gamma \qquad \Gamma \vdash \alpha \qquad \forall y, S.\ (y{:}S)^{\alpha}_{\text{glob}} \notin \Gamma}{\vdash \Gamma, (x{:}T)^{\alpha}_{\text{glob}}}\ \textit{(wf-code-var-glob)}$$

Fig. 15. Well-formed typing contexts of $\text{CC}^{\alpha,\star}_{\text{Ref}}$

$\boxed{\Gamma;\Sigma;e \vdash M^\circ : T}$

$$\frac{k : \iota}{\Gamma;\Sigma;e \vdash k : \iota}\ (\vdash \textit{code-const}) \qquad \frac{(x{:}T)^\alpha_\rho \in \Gamma \qquad \mathsf{Sub}(\Gamma) \vdash \alpha <: e}{\Gamma;\Sigma;e \vdash x : T}\ (\vdash \textit{code-var})$$

$$\frac{\alpha \notin \Gamma \qquad \Gamma, \alpha_{\mathsf{lex}}, (e{<:}\alpha)_{\mathsf{lex}}, (x{:}T_1)^\alpha_{\mathsf{lex}}; \Sigma; \alpha \vdash M^\circ : T_2}{\Gamma;\Sigma;e \vdash \lambda(x{:}T_1)^\alpha.\, M^\circ : T_1 \to T_2}\ (\vdash \textit{code-abs})$$

$$\frac{\Gamma \vdash \alpha \qquad \alpha_{\mathsf{glob}} \in \Gamma \qquad (x{:}T_1)^\alpha_{\mathsf{glob}} \in \Gamma \qquad (e_1{<:}\alpha)_{\mathsf{glob}} \in \Gamma \qquad \mathsf{Sub}(\Gamma) \vdash e_1 <: e_2 \qquad \Gamma;\Sigma;\alpha \vdash M^\circ : T_2}{\Gamma;\Sigma;e_2 \vdash \overline{\lambda}(x{:}T_1)^\alpha.\, M^\circ : T_1 \to T_2}\ (\vdash \textit{code-abs-val})$$

$$\frac{\Gamma;\Sigma;e \vdash L^\circ : T_1 \to T_2 \qquad \Gamma;\Sigma;e \vdash M^\circ : T_1}{\Gamma;\Sigma;e \vdash L^\circ\, M^\circ : T_2}\ (\vdash \textit{code-app})$$

$$\frac{\Gamma;\Sigma \vdash M : \grave{}\grave{}T\,''e}{\Gamma;\Sigma;e \vdash {\sim}M : T}\ (\vdash \textit{splice}) \qquad \frac{}{\Gamma;\Sigma;e \vdash \mathtt{blame}\ \ell : T}\ (\vdash \textit{code-blame})$$

$\boxed{\Gamma;\Sigma \vdash M : \hat{A}}$

$$\frac{k : \iota}{\Gamma;\Sigma \vdash k : \iota}\ (\vdash \textit{meta-const}) \qquad \frac{x : \hat{A} \in \Gamma}{\Gamma;\Sigma \vdash x : \hat{A}}\ (\vdash \textit{meta-var}) \qquad \frac{a : \hat{A} \in \Sigma}{\Gamma;\Sigma \vdash a : \mathtt{Ref}\ \hat{A}}\ (\vdash \textit{addr})$$

$$\frac{\Gamma \vdash \hat{A} \qquad \Gamma, x : \hat{A}; \Sigma \vdash M : \hat{B}}{\Gamma;\Sigma \vdash \lambda x.\, M : \hat{A} \to \hat{B}}\ (\vdash \textit{meta-abs}) \qquad \frac{\Gamma;\Sigma \vdash L : \hat{A} \to \hat{B} \qquad \Gamma;\Sigma \vdash M : \hat{A}}{\Gamma;\Sigma \vdash L\, M : \hat{B}}\ (\vdash \textit{meta-app})$$

$$\frac{\Gamma, \alpha_{\mathsf{lex}}; \Sigma \vdash M : \hat{A} \qquad \alpha \notin \Gamma}{\Gamma;\Sigma \vdash \Lambda\alpha.\, M : \forall\alpha.\, \hat{A}}\ (\vdash \textit{ec-abs}) \qquad \frac{\Gamma;\Sigma \vdash M : \forall\alpha.\, \hat{A} \qquad \Gamma \vdash e}{\Gamma;\Sigma \vdash M[e] : \hat{A}[\alpha := e]}\ (\vdash \textit{ec-app})$$

$$\frac{\Gamma \vdash e_1 \qquad \Gamma \vdash e_2 \qquad \Gamma, (e_1{<:}e_2)_{\mathsf{lex}}; \Sigma \vdash M : \hat{A}}{\Gamma;\Sigma \vdash e_1{<:}e_2 \Rightarrow M : e_1{<:}e_2 \Rightarrow \hat{A}}\ (\vdash \textit{sub-intro})$$

$$\frac{\Gamma;\Sigma \vdash M : e_1{<:}e_2 \Rightarrow \hat{A} \qquad \mathsf{Sub}(\Gamma) \vdash e_1 <: e_2}{\Gamma;\Sigma \vdash M\bullet : \hat{A}}\ (\vdash \textit{sub-elim})$$

$$\frac{\Gamma;\Sigma \vdash M : \star}{\Gamma;\Sigma \vdash M\bullet\star^\ell : \star}\ (\vdash \textit{sub-elim-}\star) \qquad \frac{\Gamma;\Sigma;e \vdash M^\circ : T \qquad \Gamma \vdash e}{\Gamma;\Sigma \vdash \grave{}\grave{}M^\circ\,''e : \grave{}\grave{}T\,''e}\ (\vdash \textit{quote})$$

$$\frac{\Gamma;\Sigma \vdash M : \hat{A}}{\Gamma;\Sigma \vdash \mathtt{ref}\ M : \mathtt{Ref}\ \hat{A}}\ (\vdash \textit{ref}) \qquad \frac{\Gamma;\Sigma \vdash L : \mathtt{Ref}\ \hat{A} \qquad \Gamma;\Sigma \vdash M : \hat{A}}{\Gamma;\Sigma \vdash L := M : \mathtt{Unit}}\ (\vdash \textit{assign})$$

$$\frac{\Gamma;\Sigma \vdash M : \mathtt{Ref}\ \hat{A}}{\Gamma;\Sigma \vdash !M : \hat{A}}\ (\vdash \textit{deref}) \qquad \frac{\Gamma;\Sigma \vdash M : \hat{A} \qquad \Gamma \vdash \bar{c} : \hat{A} \Rightarrow \hat{B}}{\Gamma;\Sigma \vdash M\langle\bar{c}\rangle : \hat{B}}\ (\vdash \textit{cast}) \qquad \frac{\Gamma \vdash \hat{A}}{\Gamma;\Sigma \vdash \mathtt{blame}\ \ell : \hat{A}}\ (\vdash \textit{blame})$$

Fig. 16. Typing rules for cast calculus $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$

$$\Gamma \vdash \Sigma \triangleq \forall a, \hat{A}.\ \text{if } \Sigma(a) = \hat{A} \text{ then } \Gamma \vdash \hat{A}$$

$$\Gamma; \Sigma \vdash^{H} \mu \triangleq \mathrm{dom}(\mu) = \mathrm{dom}(\Sigma) \ \wedge\ (\forall a, V, \hat{A}.\ \text{if } \mu(a) = V \text{ and } \Sigma(a) = \hat{A} \text{ then } \Gamma; \Sigma \vdash V : \hat{A})$$

$$\boxed{\Sigma \vdash^{C} C : \hat{A}} \qquad \boxed{\Sigma; e \vdash^{C} C^{\circ} : T}$$

$$\frac{\vdash \Gamma \quad \forall x, \hat{B}.\ x : \hat{B} \notin \Gamma \quad \Gamma \vdash \Sigma \quad \Gamma; \Sigma \vdash^{H} \mu \quad \Gamma; \Sigma \vdash M : \hat{A}}{\Sigma \vdash^{C} \langle \Gamma; M; \mu \rangle : \hat{A}}\ (\vdash\text{-meta-config})$$

$$\frac{\vdash \Gamma \quad \forall x, \hat{B}.\ x : \hat{B} \notin \Gamma \quad \Gamma \vdash \Sigma \quad \Gamma; \Sigma \vdash^{H} \mu \quad \Gamma; \Sigma; e \vdash M^{\circ} : T}{\Sigma; e \vdash^{C} \langle \Gamma; M^{\circ}; \mu \rangle : T}\ (\vdash\text{-code-config})$$

Fig. 17. Well-typed runtime configurations for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$

$$\boxed{M^{\circ} \longrightarrow^{\circ} N^{\circ}}$$

$$\frac{}{\sim``V^{\circ}\,''\,e_1 \longrightarrow^{\circ} V^{\circ}}\ (\textit{splice-}\beta) \qquad \frac{}{\sim((``V^{\circ}\,''\,e_1)\langle``[]\,''\,[e_1 \uparrow e_2]\rangle) \longrightarrow^{\circ} V^{\circ}}\ (\textit{splice-sub})$$

$$\frac{}{F^{\circ}[\mathtt{blame}\ \ell] \longrightarrow^{\circ} \mathtt{blame}\ \ell}\ (\xi\text{-code-blame}) \qquad \frac{}{\overline{\lambda}(x{:}T)^{\alpha}.\,\mathtt{blame}\ \ell \longrightarrow^{\circ} \mathtt{blame}\ \ell}\ (\xi\text{-code-}\overline{\lambda}\text{-blame})$$

$$\frac{}{\sim(\mathtt{blame}\ \ell) \longrightarrow^{\circ} \mathtt{blame}\ \ell}\ (\textit{splice-blame})$$

$$\boxed{e \vdash C^{\circ} \longrightarrow^{\circ} \mathcal{D}^{\circ}}$$

$$\frac{M^{\circ} \longrightarrow^{\circ} N^{\circ}}{e \vdash \langle \Gamma; M^{\circ}; \mu \rangle \longrightarrow^{\circ} \langle \Gamma; N^{\circ}; \mu \rangle}\ (\textit{pure-code})$$

$$\frac{e \vdash \langle \Gamma; M^{\circ}; \mu \rangle \longrightarrow^{\circ} \langle \Gamma'; N^{\circ}; \mu' \rangle}{e \vdash \langle \Gamma; F^{\circ}[M^{\circ}]; \mu \rangle \longrightarrow^{\circ} \langle \Gamma'; F^{\circ}[N^{\circ}]; \mu' \rangle}\ (\xi\text{-code}) \qquad \frac{\langle \Gamma; M; \mu \rangle \longrightarrow^{m} \langle \Gamma'; N; \mu' \rangle}{e \vdash \langle \Gamma; \sim M; \mu \rangle \longrightarrow^{\circ} \langle \Gamma'; \sim N; \mu' \rangle}\ (\xi\text{-splice})$$

$$\frac{\beta, y \notin \Gamma}{e \vdash \langle \Gamma; \lambda(x{:}T)^{\alpha}.\,N^{\circ}; \mu \rangle \longrightarrow^{\circ} \langle (\Gamma, \beta_{\mathrm{glob}}, (e{<:}\beta)_{\mathrm{glob}}, (y{:}T)^{\beta}_{\mathrm{glob}}); \overline{\lambda}(y{:}T)^{\beta}.\,N^{\circ}[\alpha := \beta][x := y]; \mu \rangle}\ (\textit{code-}\lambda)$$

$$\frac{\alpha \vdash \langle \Gamma; N^{\circ}; \mu \rangle \longrightarrow^{\circ} \langle \Gamma'; N^{\circ\prime}; \mu' \rangle}{e \vdash \langle \Gamma; \overline{\lambda}(x{:}T)^{\alpha}.\,N^{\circ}; \mu \rangle \longrightarrow^{\circ} \langle \Gamma'; \overline{\lambda}(x{:}T)^{\alpha}.\,N^{\circ\prime}; \mu' \rangle}\ (\xi\text{-code-}\overline{\lambda})$$

Fig. 18. Code reduction rules for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$

$\boxed{\Theta \vdash M \longrightarrow^m N}$

$$\frac{}{\Theta \vdash F[\texttt{blame}\ \ell] \longrightarrow^m \texttt{blame}\ \ell}\ (\xi\text{-meta-blame}) \qquad \frac{}{\Theta \vdash \texttt{``blame}\ \ell\ \texttt{''}\ e \longrightarrow^m \texttt{blame}\ \ell}\ (\text{quote-blame})$$

$$\frac{}{\Theta \vdash (\lambda x.\, N)\, V \longrightarrow^m N[x := V]}\ (\beta) \qquad \frac{\mathsf{Inert}([\bar{c} \to \bar{d}])}{\Theta \vdash (U\langle[\bar{c} \to \bar{d}]\rangle)\, W \longrightarrow^m (U\,(W\langle\bar{c}\rangle))\langle\bar{d}\rangle}\ (\beta\text{-cast})$$

$$\frac{}{\Theta \vdash (\Lambda\alpha.\, M)[e] \longrightarrow^m M[\alpha := e]}\ (\beta\text{-ec}) \qquad \frac{}{\Theta \vdash (U\langle[\forall\alpha.\, \bar{c}]\rangle)[e] \longrightarrow^m U[e]\langle\bar{c}[\alpha := e]\rangle}\ (\beta\text{-ec-cast})$$

$$\frac{}{\Theta \vdash (e_1 <: e_2 \Rightarrow M)\bullet \longrightarrow^m M}\ (\beta\text{-sub}) \qquad \frac{}{\Theta \vdash (U\langle[e_1 <: e_2 \Rightarrow \bar{c}]\rangle)\bullet \longrightarrow^m (U\bullet)\langle\bar{c}\rangle}\ (\beta\text{-sub-cast})$$

$$\frac{\Theta \vdash \bar{c} \longrightarrow \bar{d}}{\Theta \vdash U\langle\bar{c}\rangle \longrightarrow^m U\langle\bar{d}\rangle}\ (\text{cast-reduce}) \qquad \frac{}{\Theta \vdash U\langle\bar{c}\rangle\langle\bar{d}\rangle \longrightarrow^m U\langle\bar{c} +\!\!+ \bar{d}\rangle}\ (\text{cast-compose})$$

$$\frac{}{\Theta \vdash U\langle[]\rangle \longrightarrow^m U}\ (\text{cast-id}) \qquad \frac{}{\Theta \vdash U\langle[\bot^\ell]\rangle \longrightarrow^m \texttt{blame}\ \ell}\ (\text{cast-blame})$$

$$\frac{\Theta \vdash e_1 <: e_2 \qquad \mathsf{Inert}(\bar{c} +\!\!+ [(e_1 <: e_2 \Rightarrow \star)!])}{\Theta \vdash U\langle\bar{c} +\!\!+ [(e_1 <: e_2 \Rightarrow \star)!]\rangle\bullet\star^\ell \longrightarrow^m (U\langle\bar{c}\rangle)\bullet}\ (\text{sub-cast-success})$$

$$\frac{\Theta \nvdash e_1 <: e_2 \qquad \mathsf{Inert}(\bar{c} +\!\!+ [(e_1 <: e_2 \Rightarrow \star)!])}{\Theta \vdash U\langle\bar{c} +\!\!+ [(e_1 <: e_2 \Rightarrow \star)!]\rangle\bullet\star^\ell \longrightarrow^m \texttt{blame}\ \ell}\ (\text{sub-cast-sub-fail})$$

$$\frac{\mathsf{Inert}(\bar{c} +\!\!+ [G!]) \qquad \forall e_1\ e_2.\ G \neq (e_1 <: e_2 \Rightarrow \star)}{\Theta \vdash U\langle\bar{c} +\!\!+ [G!]\rangle\bullet\star^\ell \longrightarrow^m \texttt{blame}\ \ell}\ (\text{sub-cast-fail})$$

$$\frac{\mathsf{Inert}([\texttt{Ref}\ \bar{c}\ \bar{d}])}{\Theta \vdash !(a\langle[\texttt{Ref}\ \bar{c}\ \bar{d}]\rangle) \longrightarrow^m (!a)\langle\bar{d}\rangle}\ (\text{deref-cast}) \qquad \frac{\mathsf{Inert}([\texttt{Ref}\ \bar{c}\ \bar{d}])}{\Theta \vdash (a\langle[\texttt{Ref}\ \bar{c}\ \bar{d}]\rangle) := V \longrightarrow^m a := (V\langle\bar{c}\rangle)}\ (\text{assign-cast})$$

$\boxed{\mathcal{C} \longrightarrow^m \mathcal{D}}$

$$\frac{\mathsf{Sub}(\Gamma) \vdash M \longrightarrow^m N}{\langle\Gamma; M; \mu\rangle \longrightarrow^m \langle\Gamma; N; \mu\rangle}\ (\text{pure-meta}) \qquad \frac{\langle\Gamma; M; \mu\rangle \longrightarrow^m \langle\Gamma'; N; \mu'\rangle}{\langle\Gamma; F[M]; \mu\rangle \longrightarrow^m \langle\Gamma'; F[N]; \mu'\rangle}\ (\xi\text{-meta})$$

$$\frac{e \vdash \langle\Gamma; M^\circ; \mu\rangle \longrightarrow^\circ \langle\Gamma'; N^\circ; \mu'\rangle}{\langle\Gamma; \texttt{``}M^\circ\texttt{''}\,e; \mu\rangle \longrightarrow^m \langle\Gamma'; \texttt{``}N^\circ\texttt{''}\,e; \mu'\rangle}\ (\xi\text{-quote})$$

$$\frac{a \notin \mathrm{dom}(\mu)}{\langle\Gamma; \texttt{ref}\ V; \mu\rangle \longrightarrow^m \langle\Gamma; a; \mu[a \mapsto V]\rangle}\ (\text{ref}) \qquad \frac{\mu(a) = V}{\langle\Gamma; !a; \mu\rangle \longrightarrow^m \langle\Gamma; V; \mu\rangle}\ (\text{deref})$$

$$\frac{}{\langle\Gamma; a := V; \mu\rangle \longrightarrow^m \langle\Gamma; (); \mu[a \mapsto V]\rangle}\ (\text{assign})$$

Fig. 19. Meta reduction rules for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$

$$\boxed{\mathsf{coerce}^e(\hat{e}_1, \hat{e}_2, \ell) = \bar{c}^e}$$

$$\begin{aligned}
\mathsf{coerce}^e(\star, \star, \ell) &= [\star\uparrow\star] \\
\mathsf{coerce}^e(\star, e, \ell) &= [e?^\ell] \\
\mathsf{coerce}^e(e, \star, \ell) &= [e!] \\
\mathsf{coerce}^e(e_1, e_2, \ell) &= [e_1\uparrow e_2]
\end{aligned}$$

$$\boxed{\mathsf{coerce}^\circ(\hat{T}_1, \hat{T}_2, \ell) = \bar{c}^\circ}$$

$$\begin{aligned}
\mathsf{coerce}^\circ(\star, \star, \ell) &= [] \\
\mathsf{coerce}^\circ(\star, \iota, \ell) &= [\iota?^\ell] \\
\mathsf{coerce}^\circ(\iota, \star, \ell) &= [\iota!] \\
\mathsf{coerce}^\circ(\iota, \iota, \ell) &= [] \\
\mathsf{coerce}^\circ(\star, \hat{T}_1 \to \hat{T}_2, \ell) &= [\star \to \star?^\ell] +\!\!+ [\mathsf{coerce}^\circ(\hat{T}_1, \star, \ell) \to \mathsf{coerce}^\circ(\star, \hat{T}_2, \ell)] \\
\mathsf{coerce}^\circ(\hat{T}_1 \to \hat{T}_2, \star, \ell) &= [\mathsf{coerce}^\circ(\star, \hat{T}_1, \ell) \to \mathsf{coerce}^\circ(\hat{T}_2, \star, \ell)] +\!\!+ [\star \to \star!] \\
\mathsf{coerce}^\circ(\hat{T}_1 \to \hat{T}_2, \hat{T}_1' \to \hat{T}_2', \ell) &= [\mathsf{coerce}^\circ(\hat{T}_1', \hat{T}_1, \ell) \to \mathsf{coerce}^\circ(\hat{T}_2, \hat{T}_2', \ell)]
\end{aligned}$$

$$\boxed{\mathsf{coerce}(\hat{A}, \hat{B}, \ell) = \bar{c}}$$

$$\begin{aligned}
&\vdots \qquad \text{(base type and function cases analogous to } \mathsf{coerce}^\circ\text{)} \\
\mathsf{coerce}(\star, \mathtt{Ref}\ \hat{A}, \ell) &= [\mathtt{Ref}\ \star?^\ell] +\!\!+ [\mathtt{Ref}\ \mathsf{coerce}(\hat{A}, \star, \ell)\ \mathsf{coerce}(\star, \hat{A}, \ell)] \\
\mathsf{coerce}(\mathtt{Ref}\ \hat{A}, \star, \ell) &= [\mathtt{Ref}\ \mathsf{coerce}(\star, \hat{A}, \ell)\ \mathsf{coerce}(\hat{A}, \star, \ell)] +\!\!+ [\mathtt{Ref}\ \star!] \\
\mathsf{coerce}(\mathtt{Ref}\ \hat{A}, \mathtt{Ref}\ \hat{B}, \ell) &= [\mathtt{Ref}\ \mathsf{coerce}(\hat{B}, \hat{A}, \ell)\ \mathsf{coerce}(\hat{A}, \hat{B}, \ell)] \\
\mathsf{coerce}(\star, \grave{}\grave{}\hat{T}\,'\,'\hat{e}, \ell) &= [\grave{}\grave{}\star\,'\,'\star?^\ell] +\!\!+ [\grave{}\grave{}\mathsf{coerce}^\circ(\star, \hat{T}, \ell)\,'\,'\mathsf{coerce}^e(\star, \hat{e}, \ell)] \\
\mathsf{coerce}(\grave{}\grave{}\hat{T}\,'\,'\hat{e}, \star, \ell) &= [\grave{}\grave{}\mathsf{coerce}^\circ(\hat{T}, \star, \ell)\,'\,'\mathsf{coerce}^e(\hat{e}, \star, \ell)] +\!\!+ [\grave{}\grave{}\star\,'\,'\star!] \\
\mathsf{coerce}(\grave{}\grave{}\hat{T}_1\,'\,'\hat{e}_1, \grave{}\grave{}\hat{T}_2\,'\,'\hat{e}_2, \ell) &= [\grave{}\grave{}\mathsf{coerce}^\circ(\hat{T}_1, \hat{T}_2, \ell)\,'\,'\mathsf{coerce}^e(\hat{e}_1, \hat{e}_2, \ell)] \\
\mathsf{coerce}(\star, \forall\alpha.\hat{A}, \ell) &= [\forall\alpha.\star?^\ell] +\!\!+ [\forall\alpha.\mathsf{coerce}(\star, \hat{A}, \ell)] \\
\mathsf{coerce}(\forall\alpha.\hat{A}, \star, \ell) &= [\forall\alpha.\mathsf{coerce}(\hat{A}, \star, \ell)] +\!\!+ [\forall\alpha.\star!] \\
\mathsf{coerce}(\forall\alpha.\hat{A}, \forall\alpha.\hat{B}, \ell) &= [\forall\alpha.\mathsf{coerce}(\hat{A}, \hat{B}, \ell)] \\
\mathsf{coerce}(\star, e_1 <: e_2 \Rightarrow \hat{A}, \ell) &= [e_1 <: e_2 \Rightarrow \star?^\ell] +\!\!+ [e_1 <: e_2 \Rightarrow \mathsf{coerce}(\star, \hat{A}, \ell)] \\
\mathsf{coerce}(e_1 <: e_2 \Rightarrow \hat{A}, \star, \ell) &= [e_1 <: e_2 \Rightarrow \mathsf{coerce}(\hat{A}, \star, \ell)] +\!\!+ [e_1 <: e_2 \Rightarrow \star!] \\
\mathsf{coerce}(e_1 <: e_2 \Rightarrow \hat{A}, e_1 <: e_2 \Rightarrow \hat{B}, \ell) &= [e_1 <: e_2 \Rightarrow \mathsf{coerce}(\hat{A}, \hat{B}, \ell)]
\end{aligned}$$

Fig. 20. Generating coercions from source and target

$$\boxed{\Gamma \vdash M : \hat{A} \rightsquigarrow M}$$

$$\frac{k : \iota}{\Gamma \vdash k : \iota \rightsquigarrow k} \qquad \frac{x : \hat{A} \in \Gamma}{\Gamma \vdash x : \hat{A} \rightsquigarrow x} \qquad \frac{\Gamma \vdash \hat{A} \qquad (\Gamma, x : \hat{A}) \vdash N : \hat{B} \rightsquigarrow N'}{\Gamma \vdash \lambda(x : \hat{A})N : \hat{A} \to \hat{B} \rightsquigarrow \lambda(x : \hat{A})N'}$$

$$\frac{\Gamma \vdash L : \hat{A}_1 \to \hat{A}_2 \rightsquigarrow L' \qquad \Gamma \vdash M : \hat{B} \rightsquigarrow M'}{\Gamma \vdash (L\ M)^{\ell} : \hat{A}_2 \rightsquigarrow L'(M'\langle \mathsf{coerce}(\hat{B}, \hat{A}_1, \ell)\rangle)}$$

$$\frac{\Gamma \vdash L : \star \rightsquigarrow L' \qquad \Gamma \vdash M : \hat{A} \rightsquigarrow M'}{\Gamma \vdash (L\ M)^{\ell} : \star \rightsquigarrow (L'\langle \mathsf{coerce}(\star, (\star \to \star), \ell)\rangle)(M'\langle \mathsf{coerce}(\hat{A}, \star, \ell)\rangle)} \qquad \frac{\Gamma, \alpha_{\mathsf{lex}} \vdash N : \hat{A} \rightsquigarrow N'}{\Gamma \vdash \Lambda\alpha.N : \forall\alpha.\hat{A} \rightsquigarrow \Lambda\alpha.N'}$$

$$\frac{\Gamma \vdash M : \forall\alpha.\hat{A} \rightsquigarrow M' \qquad \Gamma \vdash e}{\Gamma \vdash M[e]^{\ell} : \hat{A}[\alpha := e] \rightsquigarrow M'[e]} \qquad \frac{\Gamma \vdash M : \star \rightsquigarrow M' \qquad \Gamma \vdash e}{\Gamma \vdash M[e]^{\ell} : \star \rightsquigarrow (M'\langle \mathsf{coerce}(\star, (\forall\beta.\star), \ell)\rangle)[e]}$$

$$\frac{\Gamma \vdash e_1 \qquad \Gamma \vdash e_2 \qquad \Gamma, (e_1 <: e_2)_{\mathsf{lex}} \vdash M : \hat{A} \rightsquigarrow M'}{\Gamma \vdash (e_1 <: e_2 \Rightarrow M) : (e_1 <: e_2 \Rightarrow \hat{A}) \rightsquigarrow (e_1 <: e_2 \Rightarrow M')}$$

$$\frac{\Gamma \vdash M : (e_1 <: e_2 \Rightarrow \hat{A}) \rightsquigarrow M' \qquad \mathsf{Sub}(\Gamma) \vdash e_1 <: e_2}{\Gamma \vdash M \bullet^{\ell} : \hat{A} \rightsquigarrow M' \bullet} \qquad \frac{\Gamma \vdash M : \star \rightsquigarrow M'}{\Gamma \vdash M \bullet^{\ell} : \star \rightsquigarrow (M' \bullet\star^{\ell})}$$

$$\frac{\Gamma; e \vdash M^{\circ} \Rightarrow T \rightsquigarrow M^{\circ\prime} \qquad \Gamma \vdash e}{\Gamma \vdash \grave{}\grave{}M^{\circ}\,\grave{}\grave{}e : \grave{}\grave{}T\,\grave{}\grave{}e \rightsquigarrow \grave{}\grave{}M^{\circ\prime}\,\grave{}\grave{}e} \qquad \frac{\Gamma \vdash M : \hat{A} \rightsquigarrow M'}{\Gamma \vdash \mathtt{ref}\ M : \mathtt{Ref}\ \hat{A} \rightsquigarrow \mathtt{ref}\ M'}$$

$$\frac{\Gamma \vdash L : \mathtt{Ref}\ \hat{A} \rightsquigarrow L' \qquad \Gamma \vdash M : \hat{B} \rightsquigarrow M'}{\Gamma \vdash L :=^{\ell} M : \mathtt{Unit} \rightsquigarrow L' := (M'\langle \mathsf{coerce}(\hat{B}, \hat{A}, \ell)\rangle)}$$

$$\frac{\Gamma \vdash L : \star \rightsquigarrow L' \qquad \Gamma \vdash M : \hat{A} \rightsquigarrow M'}{\Gamma \vdash L :=^{\ell} M : \mathtt{Unit} \rightsquigarrow (L'\langle \mathsf{coerce}(\star, (\mathtt{Ref}\ \star), \ell)\rangle) := (M'\langle \mathsf{coerce}(\hat{A}, \star, \ell)\rangle)}$$

$$\frac{\Gamma \vdash M : \mathtt{Ref}\ \hat{A} \rightsquigarrow M'}{\Gamma \vdash !\,M : \hat{A} \rightsquigarrow !\,M'} \qquad \frac{\Gamma \vdash M : \star \rightsquigarrow M'}{\Gamma \vdash !^{\ell}\,M : \star \rightsquigarrow !\,(M'\langle \mathsf{coerce}(\star, \mathtt{Ref}\ \star, \ell)\rangle)}$$

$$\boxed{\Gamma; e \vdash M^{\circ} \Rightarrow T \rightsquigarrow M^{\circ}}$$

$$\frac{(x{:}T)^{\alpha}_{\rho} \in \Gamma \qquad \mathsf{Sub}(\Gamma) \vdash \alpha <: e}{\Gamma; e \vdash x \Rightarrow T \rightsquigarrow x} \qquad \frac{\Gamma; e \vdash L^{\circ} \Rightarrow T_1 \to T_2 \rightsquigarrow L^{\circ\prime} \qquad \Gamma; e \vdash M^{\circ} \Leftarrow T_1 \rightsquigarrow M^{\circ\prime}}{\Gamma; e \vdash L^{\circ}\ M^{\circ} \Rightarrow T_2 \rightsquigarrow L^{\circ\prime}\ M^{\circ\prime}}$$

$$\frac{\Gamma, \alpha_{\mathsf{lex}}, (e <: \alpha)_{\mathsf{lex}}, (x{:}T_1)^{\alpha}_{\mathsf{lex}}; \alpha \vdash N^{\circ} \Rightarrow T_2 \rightsquigarrow N^{\circ\prime} \qquad \alpha \notin \Gamma}{\Gamma; e \vdash \lambda(x{:}T_1)^{\alpha}.N^{\circ} \Rightarrow T_1 \to T_2 \rightsquigarrow \lambda(x{:}T_1)^{\alpha}.N^{\circ\prime}}$$

$$\boxed{\Gamma; e \vdash M^{\circ} \Leftarrow T \rightsquigarrow M^{\circ}}$$

$$\frac{\Gamma \vdash M : \hat{A} \rightsquigarrow M'}{\Gamma; e \vdash {\sim^{\ell}}\,M \Leftarrow T \rightsquigarrow {\sim}M'\langle \mathsf{coerce}(\hat{A}, \grave{}\grave{}T\,\grave{}\grave{}e, \ell)\rangle} \qquad \frac{\Gamma; e \vdash M^{\circ} \Rightarrow T \rightsquigarrow M'}{\Gamma; e \vdash M^{\circ} \Leftarrow T \rightsquigarrow M'}$$

Fig. 21. Elaboration from $\lambda^{\alpha,\star}_{\mathrm{Ref}}$ to $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$

$$
\begin{array}{rcl}
\mathrm{EC}(\Gamma) & \triangleq & \text{the ordered subsequence of entries } \alpha_\rho \text{ in } \Gamma \\
\mathrm{Code}(\Gamma) & \triangleq & \text{the ordered subsequence of entries } (x{:}T)^\alpha_\rho \text{ in } \Gamma \\
\mathrm{Sub}(\Gamma) & \triangleq & \text{the ordered subsequence of entries } (e_1{<:}e_2)_\rho \text{ in } \Gamma \\
\mathrm{Ctx}(\langle \Gamma; M; \mu \rangle) & \triangleq & \Gamma \\
\mathrm{Ctx}(\langle \Gamma; M^\circ; \mu \rangle) & \triangleq & \Gamma \\
\mathrm{EC}(\Gamma_1) \subseteq \mathrm{EC}(\Gamma_2) & \triangleq & \forall \alpha, \rho.\ \text{if } \alpha_\rho \in \mathrm{EC}(\Gamma_1) \text{ then } \alpha_\rho \in \mathrm{EC}(\Gamma_2) \\
\mathrm{Code}(\Gamma_1) \subseteq \mathrm{Code}(\Gamma_2) & \triangleq & \forall x, T, \alpha, \rho.\ \text{if } (x{:}T)^\alpha_\rho \in \mathrm{Code}(\Gamma_1) \text{ then } (x{:}T)^\alpha_\rho \in \mathrm{Code}(\Gamma_2) \\
\mathrm{Sub}(\Gamma_1) \subseteq \mathrm{Sub}(\Gamma_2) & \triangleq & \forall e_1, e_2.\ \text{if } \mathrm{Sub}(\Gamma_1) \vdash e_1{<:}e_2 \text{ then } \mathrm{Sub}(\Gamma_2) \vdash e_1{<:}e_2 \\
 & & \quad \wedge\ \forall e, \alpha.\ \text{if } (e{<:}\alpha)_{\mathrm{glob}} \in \mathrm{Sub}(\Gamma_1) \text{ then } (e{<:}\alpha)_{\mathrm{glob}} \in \mathrm{Sub}(\Gamma_2) \\
\Sigma_1 \subseteq \Sigma_2 & \triangleq & \forall a, \hat{A}.\ \text{if } a : \hat{A} \in \Sigma_1 \text{ then } a : \hat{A} \in \Sigma_2 \\
\Gamma_1 \subseteq \Gamma_2 & \triangleq & \mathrm{EC}(\Gamma_1) \subseteq \mathrm{EC}(\Gamma_2) \wedge \mathrm{Code}(\Gamma_1) \subseteq \mathrm{Code}(\Gamma_2) \wedge \mathrm{Sub}(\Gamma_1) \subseteq \mathrm{Sub}(\Gamma_2).
\end{array}
$$

Fig. 22. Projection functions and inclusion relations used by preservation for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$

```
let rec sort : Array Int → Int → Int → Unit =
  λ (a : Array ⋆) (lo : Int) (hi : Int).
    if lo < hi then
      let pivot : Int = partition a lo hi in
      sort a lo (pivot - 1);
      sort a (pivot + 1) hi

let swap : Array Int → Int → Int → Unit =
  λ (a : Array Int) (i : Int) (j : Int).
    let temp : Int = a.(i) in
    a.(i) <- a.(j);
    a.(j) <- temp

let partition : Array Int → Int → Int → Int =
  λ (a : Array Int) (l : Int) (h : Int).
    let p : Int = a.(h) in
    let i : Ref Int = ref (h - 1) in
    for j = l to h do
      if a.(j) <= p then
        i := !i + 1;
        swap a !i j
    done;
    swap a (!i + 1) h;
    !i + 1
```

Fig. 23. Gradual quicksort program with potential catastrophic slowdown

meta hypercoercions $c, d ::= h; m; t$
meta head coercions $h ::= H?^{\ell} \mid \mathtt{id}$
meta middle coercions $m ::= c \to d \mid \mathtt{Ref}\ c\ d \mid \grave{}\grave{}c^{\circ}\ '\!'\ c^{e} \mid \forall\alpha.c \mid e{<:}e \Rightarrow c \mid \mathtt{id}_{\star} \mid \mathtt{id}_{\iota} \mid \bot^{\ell}$
meta tail coercions $t ::= G! \mid \mathtt{id}$

$\boxed{\Gamma \vdash c : \hat{A} \Rightarrow \hat{B}}$

$$\frac{\Gamma \vdash h : \hat{A} \Rightarrow \hat{A}' \qquad \Gamma \vdash m : \hat{A}' \Rightarrow \hat{B}' \qquad \Gamma \vdash t : \hat{B}' \Rightarrow \hat{B}}{\Gamma \vdash h; m; t : \hat{A} \Rightarrow \hat{B}}$$

$\boxed{\Theta \vdash c_1 \fatsemi c_2 = d}$

$$\begin{array}{lllll}
\Theta \vdash & h_1; \bot^{\ell}; t_1 & \fatsemi & h_2; m_2; t_2 & = h_1; \bot^{\ell}; t_2 \\
\Theta \vdash & h_1; m_1; t_1 & \fatsemi & h_2; \bot^{\ell}; t_2 & = \mathtt{match}\ t_1 \fatsemi h_2\ \mathtt{with} \\
& & & & \quad \mathtt{case}\ \bot^{\ell'} \Rightarrow h_1; \bot^{\ell'}; t_2 \\
& & & & \quad \mathtt{case}\ _ \Rightarrow h_1; \bot^{\ell}; t_2 \\
\Theta \vdash & h_1; m_1; t_1 & \fatsemi & h_2; m_2; t_2 & = \mathtt{match}\ t_1 \fatsemi h_2\ \mathtt{with} \\
& & & & \quad \mathtt{case}\ \mathtt{id} \Rightarrow h_1; (\Theta \vdash m_1 \fatsemi m_2); t_2 \\
& & & & \quad \mathtt{case}\ G! \Rightarrow h_1; m_1; G! \\
& & & & \quad \mathtt{case}\ H?^{\ell} \Rightarrow H?^{\ell}; m_2; t_2 \\
& & & & \quad \mathtt{case}\ \bot^{\ell} \Rightarrow h_1; \bot^{\ell}; t_2
\end{array}$$

$\boxed{t \fatsemi h = t \cup h \cup \bot^{\ell}}$

$$\begin{array}{rcll}
t & \fatsemi & \mathtt{id} & = t \\
\mathtt{id} & \fatsemi & h & = h \\
G! & \fatsemi & G?^{\ell} & = \mathtt{id} \\
G! & \fatsemi & H?^{\ell} & = \bot^{\ell} \quad , \text{if } G \neq H
\end{array}$$

$\boxed{\Theta \vdash m_1 \fatsemi m_2 = m}$

$$\begin{array}{lllll}
\Theta \vdash & m_1 & \fatsemi & \mathtt{id} & = m_1 \\
\Theta \vdash & \mathtt{id} & \fatsemi & m_2 & = m_2 \\
\Theta \vdash & c_1 \to d_1 & \fatsemi & c_2 \to d_2 & = (\Theta \vdash c_2 \fatsemi c_1) \to (\Theta \vdash d_1 \fatsemi d_2) \\
\Theta \vdash & \grave{}\grave{}c_1^{\circ}\ '\!'\ c_1^{e} & \fatsemi & \grave{}\grave{}c_2^{\circ}\ '\!'\ c_2^{e} & = \mathtt{match}\ (c_1^{\circ} \fatsemi c_2^{\circ})\ (\Theta \vdash c_1^{e} \fatsemi c_2^{e})\ \mathtt{with} \\
& & & & \quad \mathtt{case}\ (h^{\circ}; \bot^{\ell}; t^{\circ})\ \ _ \Rightarrow \bot^{\ell} \\
& & & & \quad \mathtt{case}\ _\ \ (h^{e}; \bot^{\ell}; t^{e}) \Rightarrow \bot^{\ell} \\
& & & & \quad \mathtt{case}\ c^{\circ}\ \ c^{e} \Rightarrow \grave{}\grave{}c^{\circ}\ '\!'\ c^{e} \\
\Theta \vdash & \mathtt{Ref}\ c_1\ d_1 & \fatsemi & \mathtt{Ref}\ c_2\ d_2 & = \mathtt{Ref}\ (\Theta \vdash c_2 \fatsemi c_1)\ (\Theta \vdash d_1 \fatsemi d_2) \\
\Theta \vdash & \forall\alpha.c & \fatsemi & \forall\alpha.d & = \forall\alpha.(\Theta \vdash c \fatsemi d) \\
\Theta \vdash & e_1{<:}e_2 \Rightarrow c & \fatsemi & e_1{<:}e_2 \Rightarrow d & = e_1{<:}e_2 \Rightarrow (\Theta, e_1{<:}e_2 \vdash c \fatsemi d) \\
\Theta \vdash & m_1 & \fatsemi & \bot^{\ell} & = \bot^{\ell} \\
\Theta \vdash & \bot^{\ell} & \fatsemi & m_2 & = \bot^{\ell}
\end{array}$$

Fig. 24. Meta hypercoercions: syntax, typing, and composition

$$
\begin{array}{lrcl}
\text{EC hypercoercions} & c^e, d^e & ::= & h^e; m^e; t^e \\
\text{EC head coercions} & h^e & ::= & e?^\ell \mid \mathtt{id} \\
\text{EC middle coercions} & m^e & ::= & e \uparrow e \mid \bot^\ell \\
\text{EC tail coercions} & t & ::= & e! \mid \mathtt{id}
\end{array}
$$

$$\boxed{\Gamma \vdash c : \hat{e_1} \Rightarrow \hat{e_2}}$$

$\vdots$ (analogous to meta hypercoercion typing)

$$\boxed{\Theta \vdash c_1^e \fatsemi c_2^e = d^e}$$

$$
\begin{array}{lllcl}
\Theta \vdash & h_1^e; \bot^\ell; t_1^e & \fatsemi & h_2^e; m_2^e; t_2^e & = & h_1^e; \bot^\ell; t_2^e \\
\Theta \vdash & h_1^e; m_1^e; t_1^e & \fatsemi & h_2^e; \bot^\ell; t_2^e & = & \text{match } t_1^e \fatsemi h_2^e \text{ with} \\
 & & & & & \quad \text{case } \bot^{\ell'} \Rightarrow h_1^e; \bot^{\ell'}; t_2^e \\
 & & & & & \quad \text{case } _ \Rightarrow h_1^e; \bot^\ell; t_2^e \\
\Theta \vdash & h_1^e; m_1^e; t_1^e & \fatsemi & h_2^e; m_2^e; t_2^e & = & \text{match } (\Theta \vdash t_1^e \fatsemi h_2^e) \text{ with} \\
 & & & & & \quad \text{case } e_1 \uparrow e_2 \Rightarrow h_1^e; (m_1^e \fatsemi (e_1 \uparrow e_2 \fatsemi m_2^e)); t_2^e \\
 & & & & & \quad \text{case } \mathtt{id} \Rightarrow h_1^e; (m_1^e \fatsemi m_2^e); t_2^e \\
 & & & & & \quad \text{case } e! \Rightarrow h_1^e; m_1^e; e! \\
 & & & & & \quad \text{case } e?^\ell \Rightarrow e?^\ell; m_2^e; t_2^e \\
 & & & & & \quad \text{case } \bot^\ell \Rightarrow h_1^e; \bot^\ell; t_2^e
\end{array}
$$

$$\boxed{\Theta \vdash t^e \fatsemi h^e = t^e \cup h^e \cup m^e}$$

$$
\begin{array}{lcll}
t^e \fatsemi \mathtt{id} & = & t^e & \\
\mathtt{id} \fatsemi h^e & = & h^e & \\
e_1! \fatsemi e_2?^\ell & = & e_1 \uparrow e_2 & , \text{if } \Theta \vdash e_1 <: e_2 \\
e_1! \fatsemi e_2?^\ell & = & \bot^\ell & , \text{if } \Theta \vdash e_1 \not<: e_2
\end{array}
$$

$$\boxed{m_1^e \fatsemi m_2^e = m^e}$$

$$
\begin{array}{lcll}
e_1 \uparrow e_2 \fatsemi e_2 \uparrow e_3 & = & e_1 \uparrow e_3 & \\
m_1^e \fatsemi \bot^\ell & = & \bot^\ell & \\
\bot^\ell \fatsemi m_2^e & = & \bot^\ell &
\end{array}
$$

Fig. 25. EC hypercoercions: syntax, typing and composition

$$
\begin{array}{lrcl}
\text{code hypercoercions} & c^\circ, d^\circ & ::= & h^\circ; m^\circ; t^\circ \\
\text{code head coercions} & h^\circ & ::= & H^\circ?^\ell \mid \mathtt{id} \\
\text{code middle coercions} & m^\circ & ::= & c^\circ \to d^\circ \mid \mathtt{id}\star \mid \mathtt{id}\,\iota \mid \bot^\ell \\
\text{code tail coercions} & t^\circ & ::= & G^\circ! \mid \mathtt{id}
\end{array}
$$

$\boxed{\vdash c : \hat{T}_1 \Rrightarrow \hat{T}_2}$

$\vdots$ (analogous to meta hypercoercion typing)

$\boxed{c_1^\circ \fatsemi c_2^\circ = d^\circ}$

$\vdots$ (analogous to meta hypercoercion composition)

$\boxed{t^\circ \fatsemi h^\circ = t^\circ \cup h^\circ \cup \bot^\ell}$

$\vdots$ (analogous to meta head-tail coercion composition)

$\boxed{m_1^\circ \fatsemi m_2^\circ = m^\circ}$

$$
\begin{array}{lcll}
m_1^\circ \fatsemi \mathtt{id} & = & m_1^\circ & \\
\mathtt{id} \fatsemi m_2^\circ & = & m_2^\circ & \\
c_1^\circ \to d_1^\circ \fatsemi c_2^\circ \to d_2^\circ & = & \multicolumn{2}{l}{\text{match } (c_2^\circ \fatsemi c_1^\circ)\ (d_1^\circ \fatsemi d_2^\circ) \text{ with}} \\
 & & \text{case} \quad (h^\circ; \bot^\ell; t^\circ) \quad _ & \Rrightarrow \bot^\ell \\
 & & \text{case} \quad _ \quad (h^\circ; \bot^\ell; t^\circ) & \Rrightarrow \bot^\ell \\
 & & \text{case} \quad c^\circ \quad d^\circ & \Rrightarrow c^\circ \to d^\circ \\
m_1^\circ \fatsemi \bot^\ell & = & \bot^\ell & \\
\bot^\ell \fatsemi m_2^\circ & = & \bot^\ell &
\end{array}
$$

Fig. 26. Code hypercoercions: syntax, typing and composition

$$\boxed{\mathsf{coerce}^e(\hat{e}_1, \hat{e}_2, \ell) = c^e}$$

$$\begin{aligned}
\mathsf{coerce}^e(\star, \star, \ell) &= \mathtt{id}; \star\uparrow\star; \mathtt{id} \\
\mathsf{coerce}^e(\star, e, \ell) &= e?^{\ell}; e\uparrow e; \mathtt{id} \\
\mathsf{coerce}^e(e, \star, \ell) &= \mathtt{id}; e\uparrow e; e! \\
\mathsf{coerce}^e(e_1, e_2, \ell) &= \mathtt{id}; e_1\uparrow e_2; \mathtt{id}
\end{aligned}$$

$$\boxed{\mathsf{coerce}^\circ(\hat{T}_1, \hat{T}_2, \ell) = c^\circ}$$

$$\begin{aligned}
\mathsf{coerce}^\circ(\star, \star, \ell) &= \mathtt{id}; \mathtt{id}; \mathtt{id} \\
\mathsf{coerce}^\circ(\star, \iota, \ell) &= \iota?^{\ell}; \mathtt{id}; \mathtt{id} \\
\mathsf{coerce}^\circ(\iota, \star, \ell) &= \mathtt{id}; \mathtt{id}; \iota! \\
\mathsf{coerce}^\circ(\iota, \iota, \ell) &= \mathtt{id}; \mathtt{id}; \mathtt{id} \\
\mathsf{coerce}^\circ(\star, \hat{T}_1 \to \hat{T}_2, \ell) &= \star\to\star?^{\ell}; \mathsf{coerce}^\circ(\hat{T}_1, \star, \ell) \to \mathsf{coerce}^\circ(\star, \hat{T}_2, \ell); \mathtt{id} \\
\mathsf{coerce}^\circ(\hat{T}_1 \to \hat{T}_2, \star, \ell) &= \mathtt{id}; \mathsf{coerce}^\circ(\star, \hat{T}_1, \ell) \to \mathsf{coerce}^\circ(\hat{T}_2, \star, \ell); \star\to\star! \\
\mathsf{coerce}^\circ(\hat{T}_1 \to \hat{T}_2, \hat{T}_1' \to \hat{T}_2', \ell) &= \mathtt{id}; \mathsf{coerce}^\circ(\hat{T}_1', \hat{T}_1, \ell) \to \mathsf{coerce}^\circ(\hat{T}_2, \hat{T}_2', \ell); \mathtt{id}
\end{aligned}$$

$$\boxed{\mathsf{coerce}(\hat{A}, \hat{B}, \ell) = c}$$

$$\begin{aligned}
&\vdots \qquad \text{(base type and function cases analogous to } \mathsf{coerce}^\circ\text{)} \\
\mathsf{coerce}(\star, \mathtt{Ref}\ \hat{A}, \ell) &= \mathtt{Ref}\ \star?^{\ell}; \mathtt{Ref}\ \mathsf{coerce}(\hat{A}, \star, \ell)\ \mathsf{coerce}(\star, \hat{A}, \ell); \mathtt{id} \\
\mathsf{coerce}(\mathtt{Ref}\ \hat{A}, \star, \ell) &= \mathtt{id}; \mathtt{Ref}\ \mathsf{coerce}(\star, \hat{A}, \ell)\ \mathsf{coerce}(\hat{A}, \star, \ell); \mathtt{Ref}\ \star! \\
\mathsf{coerce}(\mathtt{Ref}\ \hat{A}, \mathtt{Ref}\ \hat{B}, \ell) &= \mathtt{id}; \mathtt{Ref}\ \mathsf{coerce}(\hat{B}, \hat{A}, \ell)\ \mathsf{coerce}(\hat{A}, \hat{B}, \ell); \mathtt{id} \\
\mathsf{coerce}(\star, \text{``}\hat{T}\text{''}\hat{e}, \ell) &= \text{``}\star\text{''}\star?^{\ell}; \text{``}\mathsf{coerce}^\circ(\star, \hat{T}, \ell)\text{''}\mathsf{coerce}^e(\star, \hat{e}, \ell); \mathtt{id} \\
\mathsf{coerce}(\text{``}\hat{T}\text{''}\hat{e}, \star, \ell) &= \mathtt{id}; \text{``}\mathsf{coerce}^\circ(\hat{T}, \star, \ell)\text{''}\mathsf{coerce}^e(\hat{e}, \star, \ell); \text{``}\star\text{''}\star! \\
\mathsf{coerce}(\text{``}\hat{T}_1\text{''}\hat{e}_1, \text{``}\hat{T}_2\text{''}\hat{e}_2, \ell) &= \mathtt{id}; \text{``}\mathsf{coerce}^\circ(\hat{T}_1, \hat{T}_2, \ell)\text{''}\mathsf{coerce}^e(\hat{e}_1, \hat{e}_2, \ell); \mathtt{id} \\
\mathsf{coerce}(\star, \forall\alpha.\hat{A}, \ell) &= \forall\alpha.\star?^{\ell}; \forall\alpha.\mathsf{coerce}(\star, \hat{A}, \ell); \mathtt{id} \\
\mathsf{coerce}(\forall\alpha.\hat{A}, \star, \ell) &= \mathtt{id}; \forall\alpha.\mathsf{coerce}(\hat{A}, \star, \ell); \forall\alpha.\star! \\
\mathsf{coerce}(\forall\alpha.\hat{A}, \forall\alpha.\hat{B}, \ell) &= \mathtt{id}; \forall\alpha.\mathsf{coerce}(\hat{A}, \hat{B}, \ell); \mathtt{id} \\
\mathsf{coerce}(\star, e_1 <: e_2 \Rightarrow \hat{A}, \ell) &= e_1 <: e_2 \Rightarrow \star?^{\ell}; e_1 <: e_2 \Rightarrow \mathsf{coerce}(\star, \hat{A}, \ell); \mathtt{id} \\
\mathsf{coerce}(e_1 <: e_2 \Rightarrow \hat{A}, \star, \ell) &= \mathtt{id}; e_1 <: e_2 \Rightarrow \mathsf{coerce}(\hat{A}, \star, \ell); e_1 <: e_2 \Rightarrow \star! \\
\mathsf{coerce}(e_1 <: e_2 \Rightarrow \hat{A}, e_1 <: e_2 \Rightarrow \hat{B}, \ell) &= \mathtt{id}; e_1 <: e_2 \Rightarrow \mathsf{coerce}(\hat{A}, \hat{B}, \ell); \mathtt{id}
\end{aligned}$$

Fig. 27. Generating hypercoercions from source and target

Updated syntax for V, F

$$\begin{array}{lll} \text{meta values} & V, W & ::= U \mid U\langle c_i \rangle \\ \text{meta frame} & F & ::= \Box\ M \mid V\ \Box \mid \mathtt{ref}\ \Box \mid !\Box \mid \Box := M \mid V := \Box \mid \Box[e] \mid \Box \bullet \mid \Box \bullet\star^{\ell} \\ \text{inert coercions} & c_i & ::= (\mathtt{id}; m_1; G!)^{\dagger_1} \mid (\mathtt{id}; m_2; \mathtt{id})^{\dagger_2} \end{array}$$

$$\dagger_1 \neg \mathsf{Blame}(m_1) \qquad \dagger_2 \neg(\mathsf{Id}(m_2) \wedge \mathsf{Blame}(m_2))$$

$$\mathsf{Id}(m) \triangleq (m = \mathtt{id}\ \star) \vee (m = \mathtt{id}\ \iota) \qquad \mathsf{Blame}(m) \triangleq \exists \ell.\ m = \bot^{\ell}$$

$$\mathrm{ctx} ::= \mathrm{Any} \mid \mathrm{NonCast}$$

$M \longrightarrow^{\mathrm{A}} N$

$$\begin{array}{rlrl} (\lambda(x{:}A)\ N)V & \longrightarrow^{\mathrm{A}} N[x := V] & (U\langle \mathtt{id}; c \to d; \mathtt{id}\rangle)\ W & \longrightarrow^{\mathrm{A}} (U\ (W\langle c\rangle))\langle d\rangle \\ (\Lambda\alpha.N)[e] & \longrightarrow^{\mathrm{A}} N[\alpha := e] & U\langle \mathtt{id}; \mathtt{Ref}\ c\ d; \mathtt{id}\rangle := V & \longrightarrow^{\mathrm{A}} U := (V\langle c\rangle) \\ (e_1{<:}e_2 \Rightarrow N) \bullet & \longrightarrow^{\mathrm{A}} N & !(U\langle \mathtt{id}; \mathtt{Ref}\ c\ d; \mathtt{id}\rangle) & \longrightarrow^{\mathrm{A}} (!U)\langle d\rangle \\ F[\mathtt{blame}\ \ell] & \longrightarrow^{\mathrm{A}} \mathtt{blame}\ \ell & U\langle \mathtt{id}; \forall\alpha.c; \mathtt{id}\rangle[e] & \longrightarrow^{\mathrm{A}} U[e]\langle c[\alpha := e]\rangle \\ ``\sim(\mathtt{blame}\ \ell)\ ''e & \longrightarrow^{\mathrm{A}} \mathtt{blame}\ \ell & U\langle \mathtt{id}; e_1{<:}e_2 \Rightarrow c; \mathtt{id}\rangle \bullet & \longrightarrow^{\mathrm{A}} (U \bullet)\langle c\rangle \end{array}$$

$M \longrightarrow^{\mathrm{NC}} N$

$$U\langle \mathtt{id}; \mathtt{id}; \mathtt{id}\rangle \longrightarrow^{\mathrm{NC}} U \qquad U\langle \mathtt{id}; \bot^{\ell}; t\rangle \longrightarrow^{\mathrm{NC}} \mathtt{blame}\ \ell$$

$\mathrm{ctx} \vdash C \longrightarrow^{m} \mathcal{D}$

$$\begin{array}{rlrl} \mathrm{NonCast} \vdash \langle \Gamma; M\langle c\rangle\langle d\rangle; \mu\rangle & \longrightarrow^{m} \langle \Gamma; M\langle \mathsf{Sub}(\Gamma) \vdash c \fatsemi d\rangle; \mu\rangle & & \\ \mathrm{Any} \vdash \langle \Gamma_1; ``M^{\circ}\ ''e; \mu_1\rangle & \longrightarrow^{m} \langle \Gamma_2; ``N^{\circ}\ ''e; \mu_2\rangle & \text{if } \langle \Gamma_1; M^{\circ}; \mu_1\rangle \mid e \longrightarrow^{\circ} \langle \Gamma_2; N^{\circ}; \mu_2\rangle & \\ \mathrm{Any} \vdash \langle \Gamma; ``V^{\circ}\ ''e; \mu\rangle & \longrightarrow^{m} \langle \Gamma; (`V^{\circ}\ `e); \mu\rangle & & \\ \mathrm{NonCast} \vdash \langle \Gamma; M; \mu\rangle & \longrightarrow^{m} \langle \Gamma; N; \mu\rangle & \text{if } M \longrightarrow^{\mathrm{NC}} N & \\ \mathrm{Any} \vdash \langle \Gamma; M; \mu\rangle & \longrightarrow^{m} \langle \Gamma; N; \mu\rangle & \text{if } M \longrightarrow^{\mathrm{A}} N & \\ \mathrm{Any} \vdash \langle \Gamma; \mathtt{ref}\ V; \mu\rangle & \longrightarrow^{m} \langle \Gamma; a; \mu[a \mapsto V]\rangle & \text{if } a \notin \mathrm{dom}(\mu) & \\ \mathrm{Any} \vdash \langle \Gamma; a := V; \mu\rangle & \longrightarrow^{m} \langle \Gamma; (); \mu[a \mapsto V]\rangle & & \\ \mathrm{Any} \vdash \langle \Gamma; !a; \mu\rangle & \longrightarrow^{m} \langle \Gamma; \mu(a); \mu\rangle & & \\ \mathrm{Any} \vdash \langle \Gamma; F[M]; \mu\rangle & \longrightarrow^{m} \langle \Gamma'; F[N]; \mu'\rangle & \text{if } \mathrm{ctx} \vdash \langle \Gamma; M; \mu\rangle \longrightarrow^{m} \langle \Gamma'; N; \mu'\rangle & \\ \mathrm{NonCast} \vdash \langle \Gamma; M\langle c\rangle; \mu\rangle & \longrightarrow^{m} \langle \Gamma'; N\langle c\rangle; \mu'\rangle & \text{if } \mathrm{Any} \vdash \langle \Gamma; M; \mu\rangle \longrightarrow^{m} \langle \Gamma'; N; \mu'\rangle & \\ \mathrm{Any} \vdash \langle \Gamma; U\langle \mathtt{id}; m; (e_1{<:}e_2 \Rightarrow \star)!\rangle \bullet \star^{\ell}; \mu\rangle & \longrightarrow^{m} \langle \Gamma; U\langle \mathtt{id}; m; \mathtt{id}\rangle \bullet; \mu\rangle & \text{if } \Gamma \vdash e_1{<:}e_2 & \\ \mathrm{Any} \vdash \langle \Gamma; U\langle \mathtt{id}; m; (e_1{<:}e_2 \Rightarrow \star)!\rangle \bullet \star^{\ell}; \mu\rangle & \longrightarrow^{m} \langle \Gamma; \mathtt{blame}\ \ell; \mu\rangle & \text{if } \Gamma \vdash e_1 \not<: e_2 & \\ \mathrm{Any} \vdash \langle \Gamma; U\langle \mathtt{id}; m; G!\rangle \bullet \star\ell; \mu\rangle & \longrightarrow^{m} \langle \Gamma; \mathtt{blame}\ \ell; \mu\rangle & \text{if } \forall e_1 e_2.\ G \neq (e_1{<:}e_2 \Rightarrow \star) & \end{array}$$

$M^{\circ} \longrightarrow^{\circ} N^{\circ}$

$$\frac{}{\sim((`V^{\circ}\ `e_1)\langle \mathtt{id}; ``\mathtt{id}; m^{\circ}; \mathtt{id}\ ''(\mathtt{id}; e_1 \uparrow e_2; \mathtt{id}); \mathtt{id}\rangle) \longrightarrow^{\circ} V^{\circ}}\ (\textit{splice-sub})$$

$\vdots$ (remaining rules same as Figure 18)

$e \vdash C^{\circ} \longrightarrow^{\circ} \mathcal{D}^{\circ}$

$$\frac{\mathrm{ctx} \vdash \langle \Gamma; M; \mu\rangle \longrightarrow^{m} \langle \Gamma'; N; \mu'\rangle}{e \vdash \langle \Gamma; \sim M; \mu\rangle \longrightarrow^{\circ} \langle \Gamma'; \sim N; \mu'\rangle}\ (\textit{ξ-splice})$$

$\vdots$ (remaining rules same as Figure 18)

Fig. 28. Space-efficient reduction rules for $\mathrm{CC}^{\alpha,\star}_{\mathrm{Ref}}$ using hypercoercions

$$d \in \mathbb{B} \qquad n \in \mathbb{N}$$

$\boxed{d \vdash^\circ M^\circ \text{ ok}}$

$$\frac{n \mid d \vdash^m M \text{ ok}}{d \vdash^\circ {\sim}M \text{ ok}}\ \textit{(ok-splice)} \qquad \frac{}{d \vdash^\circ k \text{ ok}}\ \textit{(ok-code-const)} \qquad \frac{}{d \vdash^\circ x \text{ ok}}\ \textit{(ok-code-var)}$$

$$\frac{d \vdash^\circ N^\circ \text{ ok}}{d \vdash^\circ \lambda(x{:}T)^\alpha.\, N^\circ \text{ ok}}\ \textit{(ok-code-abs)} \qquad \frac{d \vdash^\circ N^\circ \text{ ok}}{d \vdash^\circ \overline{\lambda}(x{:}T)^\alpha.\, N^\circ \text{ ok}}\ \textit{(ok-code-abs-val)}$$

$$\frac{d \vdash^\circ L^\circ \text{ ok} \qquad d \vdash^\circ M^\circ \text{ ok}}{d \vdash^\circ L^\circ\, M^\circ \text{ ok}}\ \textit{(ok-code-app)} \qquad \frac{}{d \vdash^\circ \texttt{blame}\ \ell \text{ ok}}\ \textit{(ok-code-blame)}$$

$\boxed{n \mid d \vdash^m M \text{ ok}}$

$$\frac{n \mid \text{true} \vdash^m M \text{ ok} \qquad n \le 1}{n+1 \mid \text{true} \vdash^m M\langle c\rangle \text{ ok}}\ \textit{(ok-cast-delayed)} \qquad \frac{n \mid \text{false} \vdash^m M \text{ ok} \qquad n \le 2}{n+1 \mid \text{false} \vdash^m M\langle c\rangle \text{ ok}}\ \textit{(ok-cast)}$$

$$\frac{}{0 \mid d \vdash^m k \text{ ok}}\ \textit{(ok-meta-const)} \qquad \frac{}{1 \mid d \vdash^m x \text{ ok}}\ \textit{(ok-meta-var)} \qquad \frac{}{0 \mid d \vdash^m a \text{ ok}}\ \textit{(ok-addr)}$$

$$\frac{n \mid \text{true} \vdash^m N \text{ ok}}{0 \mid d \vdash^m \lambda x.\, N \text{ ok}}\ \textit{(ok-meta-abs)} \qquad \frac{n \mid d \vdash^m L \text{ ok} \qquad n' \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m L\, M \text{ ok}}\ \textit{(ok-meta-app)}$$

$$\frac{n \mid \text{true} \vdash^m N \text{ ok}}{0 \mid d \vdash^m \Lambda\alpha.\, N \text{ ok}}\ \textit{(ok-ec-abs)} \qquad \frac{n \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m M[e] \text{ ok}}\ \textit{(ok-ec-app)}$$

$$\frac{n \mid \text{true} \vdash^m N \text{ ok}}{0 \mid d \vdash^m e_1 <: e_2 \Rightarrow N \text{ ok}}\ \textit{(ok-sub-intro)} \qquad \frac{n \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m M\bullet \text{ ok}}\ \textit{(ok-sub-elim)}$$

$$\frac{n \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m M\bullet\star^\ell \text{ ok}}\ \textit{(ok-sub-elim-}\star\textit{)} \qquad \frac{d \vdash^\circ M^\circ \text{ ok}}{0 \mid d \vdash^m \grave{}\grave{}M^\circ{}'' e \text{ ok}}\ \textit{(ok-quote)}$$

$$\frac{n \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m \texttt{ref}\ M \text{ ok}}\ \textit{(ok-ref)} \qquad \frac{n \mid d \vdash^m L \text{ ok} \qquad n' \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m L := M \text{ ok}}\ \textit{(ok-assign)}$$

$$\frac{n \mid d \vdash^m M \text{ ok}}{0 \mid d \vdash^m !M \text{ ok}}\ \textit{(ok-deref)} \qquad \frac{}{0 \mid d \vdash^m \texttt{blame}\ \ell \text{ ok}}\ \textit{(ok-meta-blame)}$$

Fig. 29. Adjacent cast judgments for code and meta terms

$\boxed{\mu \text{ ok}}$

$$\frac{}{\emptyset \text{ ok}}\ \textit{(}\mu\textit{ok-empty)} \qquad \frac{n \mid \text{false} \vdash^m V \text{ ok} \qquad \mu \text{ ok}}{(\mu[a \mapsto V]) \text{ ok}}\ \textit{(}\mu\textit{ok-cons)}$$

Fig. 30. Adjacent cast judgment for heaps

$$\boxed{\mathsf{height} : c \cup c^\circ \to \mathbb{N}}$$

$$\begin{aligned} \mathsf{height}(h^o; m^o; t^o) &= \mathsf{height}(m^o) \\ \mathsf{height}(h; m; t) &= \mathsf{height}(m) \end{aligned}$$

$$\boxed{\mathsf{height} : c^e \to \mathbb{N}}$$

$$\mathsf{height}(c^e) = 1$$

$$\boxed{\mathsf{height} : m \cup m^\circ \to \mathbb{N}}$$

$$\begin{array}{lll} \mathsf{height}(\mathtt{id}) & = \mathsf{height}(\bot^\ell) & = 1 \\ \mathsf{height}(c_1 \to c_2) & = \mathsf{height}(\mathtt{Ref}\ c_1\ c_2) & = 1 + \max(\mathsf{height}(c_1), \mathsf{height}(c_2)) \\ \mathsf{height}(c_1^\circ \to c_2^\circ) & & = 1 + \max(\mathsf{height}(c_1^\circ), \mathsf{height}(c_2^\circ)) \\ \mathsf{height}(\grave{}\grave{}\, c^\circ\, '' c^e) & & = 1 + \max(\mathsf{height}(c^\circ), \mathsf{height}(c^e)) \\ \mathsf{height}(\forall \alpha.c) & = \mathsf{height}(e_1 <: e_2 \Rightarrow c) & = 1 + \mathsf{height}(c) \end{array}$$

Fig. 31. Hypercoercion height

$$\boxed{\mathsf{size} : c \cup c^\circ \to \mathbb{N}}$$

$$\begin{aligned} \mathsf{size}(h^o; m^o; t^o) &= 4 + \mathsf{size}(m^o) \\ \mathsf{size}(h; m; t) &= 4 + \mathsf{size}(m) \end{aligned}$$

$$\boxed{\mathsf{size} : c^e \to \mathbb{N}}$$

$$\mathsf{size}(c^e) = 5$$

$$\boxed{\mathsf{size} : m \cup m^\circ \to \mathbb{N}}$$

$$\begin{array}{lll} \mathsf{size}(\mathtt{id}) & = \mathsf{size}(\bot^\ell) & = 1 \\ \mathsf{size}(c_1 \to c_2) & = \mathsf{size}(\mathtt{Ref}\ c_1\ c_2) & = 1 + \mathsf{size}(c_1) + \mathsf{size}(c_2) \\ \mathsf{size}(c_1^\circ \to c_2^\circ) & & = 1 + \mathsf{size}(c_1^\circ) + \mathsf{size}(c_2^\circ) \\ \mathsf{size}(\grave{}\grave{}\, c^\circ\, '' c^e) & & = 1 + \mathsf{size}(c^\circ) + \mathsf{size}(c^e) \\ \mathsf{size}(\forall \alpha.c) & = \mathsf{size}(e_1 <: e_2 \Rightarrow c) & = 1 + \mathsf{size}(c) \end{array}$$

Fig. 32. Hypercoercion size

$$\boxed{\mathsf{c_height} : M \cup M^\circ \to \mathbb{N}}$$

$$\begin{aligned}
\mathsf{c_height}(k) &= 0 \\
\mathsf{c_height}(x) &= 0 \\
\mathsf{c_height}(a) &= 0 \\
\mathsf{c_height}(\texttt{blame}\ \ell) &= 0 \\
\mathsf{c_height}(\lambda(x{:}T)^\alpha.\, N^\circ) &= \mathsf{c_height}(N^\circ) \\
\mathsf{c_height}(\overline{\lambda}(x{:}T)^\alpha.\, N^\circ) &= \mathsf{c_height}(N^\circ) \\
\mathsf{c_height}(L^\circ\ M^\circ) &= \max(\mathsf{c_height}(L^\circ), \mathsf{c_height}(M^\circ)) \\
\mathsf{c_height}(\sim M) &= \mathsf{c_height}(M) \\
\mathsf{c_height}(\lambda(x : \hat{A}).\, N) &= \mathsf{c_height}(N) \\
\mathsf{c_height}(\Lambda\alpha.\, N) &= \mathsf{c_height}(N) \\
\mathsf{c_height}(e_1 <: e_2 \Rightarrow N) &= \mathsf{c_height}(N) \\
\mathsf{c_height}(\grave{}\grave{}M^\circ\,'\,'\,e) &= \mathsf{c_height}(M^\circ) \\
\mathsf{c_height}(L\ M) &= \max(\mathsf{c_height}(L), \mathsf{c_height}(M)) \\
\mathsf{c_height}(L := M) &= \max(\mathsf{c_height}(L), \mathsf{c_height}(M)) \\
\mathsf{c_height}(M[e]) &= \mathsf{c_height}(M) \\
\mathsf{c_height}(M\bullet) &= \mathsf{c_height}(M) \\
\mathsf{c_height}(M\bullet\star^\ell) &= \mathsf{c_height}(M) \\
\mathsf{c_height}(\texttt{ref}\ M) &= \mathsf{c_height}(M) \\
\mathsf{c_height}(!\,M) &= \mathsf{c_height}(M) \\
\mathsf{c_height}(M\langle c\rangle) &= \max(\mathsf{c_height}(M), \mathsf{height}(c))
\end{aligned}$$

Fig. 33. Cast height for code and meta terms

$$\boxed{\mathsf{size} : M \cup M^\circ \to \mathbb{N}}$$

$$\begin{aligned}
\mathsf{size}(k) = \mathsf{size}(x) = \mathsf{size}(\texttt{blame}\ \ell) = \mathsf{size}(a) &= 1 \\
\mathsf{size}(\lambda(x{:}T)^\alpha.\, N^\circ) = \mathsf{size}(\overline{\lambda}(x{:}T)^\alpha.\, N^\circ) &= 1 + \mathsf{size}^\circ(N^\circ) \\
\mathsf{size}^\circ(L^\circ\ M^\circ) &= 1 + \mathsf{size}(L^\circ) + \mathsf{size}(M^\circ) \\
\mathsf{size}(\sim M) &= 1 + \mathsf{size}(M) \\
\mathsf{size}(\lambda(x : \hat{A}).\, N) = \mathsf{size}(\Lambda\alpha.\, N) = \mathsf{size}(e_1 <: e_2 \Rightarrow N) &= 1 + \mathsf{size}(N) \\
\mathsf{size}(L\ M) = \mathsf{size}(L := M) &= 1 + \mathsf{size}(L) + \mathsf{size}(M) \\
\mathsf{size}(M[e]) = \mathsf{size}(M\bullet) = \mathsf{size}(M\bullet\star^\ell) = \mathsf{size}(\texttt{ref}\ M) = \mathsf{size}(!\,M) &= 1 + \mathsf{size}(M) \\
\mathsf{size}(\grave{}\grave{}M^\circ\,'\,'\,e) &= 1 + \mathsf{size}(M^\circ) \\
\mathsf{size}(M\langle c\rangle) &= \mathsf{size}(M) + \mathsf{size}(c)
\end{aligned}$$

$$\boxed{\mathsf{num} : M \cup M^\circ \to \mathbb{N}}$$

$$\begin{aligned}
&\vdots \qquad \text{(remaining cases are analogous to size)} \\
\mathsf{num}(M\langle c\rangle) &= \mathsf{num}(M) + 1
\end{aligned}$$

$$\boxed{\mathsf{ideal_size} : M \cup M^\circ \to \mathbb{N}}$$

$$\begin{aligned}
&\vdots \qquad \text{(remaining cases are analogous to size)} \\
\mathsf{ideal_size}(M\langle c\rangle) &= \mathsf{ideal_size}(M)
\end{aligned}$$

Fig. 34. size, num, and ideal_size for code and meta terms

$\boxed{\mathsf{c_height}, \mathsf{size}, \mathsf{ideal_size} : \mu \to \mathbb{N}}$

$$\begin{array}{rcl@{\qquad}rcl}
\mathsf{c_height}(\emptyset) & = & 0 & \mathsf{c_height}(\mu[a \mapsto V]) & = & \max(\mathsf{c_height}(V), \mathsf{c_height}(\mu)) \\
\mathsf{size}(\emptyset) & = & 0 & \mathsf{size}(\mu[a \mapsto V]) & = & \mathsf{size}(V) + \mathsf{size}(\mu) \\
\mathsf{ideal_size}(\emptyset) & = & 0 & \mathsf{ideal_size}(\mu[a \mapsto V]) & = & \mathsf{ideal_size}(V) + \mathsf{ideal_size}(\mu)
\end{array}$$

Fig. 35. Sizes and height functions for heaps

$\boxed{\mathsf{c_height}, \mathsf{size}, \mathsf{ideal_size} : C \cup C^\circ \to \mathbb{N}}$

$$\begin{array}{rcl}
\mathsf{c_height}(\langle \Gamma; M; \mu \rangle) & = & \max(\mathsf{c_height}(\mu), \mathsf{c_height}(M)) \\
\mathsf{size}(\langle \Gamma; M; \mu \rangle) & = & \mathsf{size}(\mu) + \mathsf{size}(M) \\
\mathsf{ideal_size}(\langle \Gamma; M; \mu \rangle) & = & \mathsf{ideal_size}(\mu) + \mathsf{ideal_size}(M)
\end{array}$$

Fig. 36. Sizes and height functions for configurations